MULTIMODAL FUSION FOR DETECTION OF COLORECTAL CANCER USING CLINICAL BERT AND PHIKONV2

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Declaration

MULTIMODAL FUSION FOR DETECTION OF COLORECTAL CANCER USING CLINICAL BERT AND PHIKONV2

The project work is submitted in partial fulfilment of academic requirements for the award of B. Tech. Degree in the Department of Computer Science Engineering of the Faculty of Engineering and Technology of M. S. Ramaiah University of Applied Sciences. The project report submitted herewith is the result of original work and conforms to the guidelines on plagiarism laid out in the University Student Handbook. All sections of the text and results obtained from other sources are fully referenced. Cheating and plagiarism constitute a breach of university regulations; accordingly, this project report has been subjected to a plagiarism check and submitted to the supervisor.

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The values of dedication, discipline, work ethics, and patience gained during this period will guide us till the end.

Abstract

Work and Motivation:

Microsatellite Instability (MSI) is a critical genomic marker in colorectal cancer which is linked to high sensitivity to immune checkpoint inhibitors (ICIs) like Pembrolizumab. Although they are accurate, conventional detection using PCR, immunohistochemistry or next-generation sequencing is slowed by the need for infrastructure and takes up to seven to fourteen days to give a result, delaying therapy. This project aims to address the unmet need for fast and easy MSI pre-screening from standard diagnostic samples.

Scope and Methodology:

The authors introduce a multimodal artificial intelligence solution that combines the information contained in Hematoxylin and Eosin stained whole-slide histopathology images with the information contained in unstructured clinical pathology reports. The visual branch uses a DINOv2-based ViT-L/16 pathology foundation model to project tissue areas into 1024-dimensional feature sequences. A pretrained model for clinical notes, ClinicalBERT, is applied to the text branch to obtain 768-dimensional embeddings for each token in the context. The main contribution is the Summary To Patch Cross Attention architecture, which maps both modalities into a shared latent space of 1024 dimensions, and implements multi-head cross-attention with clinical text queries.

Results and Conclusions:

The system was trained using a weighted binary cross-entropy loss function, where the delay in convergence to a plateau of 4.17 was done deliberately, and a conservative decision threshold of 0.83 was used to achieve a high precision of MSI. The validation cohort was an isolated group of 66 patients, with 89.39% overall accuracy, 0.9150 ROC-AUC, and 0.80 MSI precision when evaluated. A dual tier Explainable AI stack and an LLM-based diagnostic scribe using Qwen 2.5 via Ollama deployed through a Streamlit based clinical user interface.

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Nomenclature

- V: Visual patch embedding sequence, RN×1024.
- T: Clinical text token, RL x 768
- T': Embedded sequence of text projected, RL×1024
- Q: The query matrix is the cross-attention matrix T’ (derived from Q).
- K: key matrix in cross-attention (from V)
- Vv: value matrix in cross attention part which is related to the value matrix V.
- -dk: scaling factor (shared latent dimension = 1024)
- ai: Attention score of i'th tissue patch in MIL
- w+: Positvie class weight in weighted BCE loss
- y^: Model predicted MSI probability (∈ [0,1])
- Th: Decision threshold (0.83 and 0.70 in the deployed system)

Abbreviation and Acronyms

- BCE: Binary Cross-Entropy
- CLS: Classification Token
- CNN: Convolutional Neural Network
- CRC: Colorectal Cancer
- DINO: Self-Distillation with No Labels
- EMA: Exponential Moving Average
- FPO: Floating Point Operations:
- GNN: Graph Neural Network
- H and E: Haematoxylin and Eosin staining.
- IHC: Immunohistochemistry
- MIL: Multiple Instance Learning
- MLP: Multi-Layer Perceptron
- MRI: Magnetic Resonance Imaging
- MSI: Microsatellite Instability
- MSS: Microsatellite Stable
- PCAM: Patch Camelyon (Dataset)
- ReLU is a ReLU.ReLU is a "ReLU"
- ROC: Receiver Operating Characteristic
- ViT: Vision Transformer
- WSI: Whole Slide Image
- XAI: Explainable Artificial IntelligenceXAI

Self- Assessment of the Project

PO/PSO	Contribution from the project	Level
Engineering Knowledge: Knowledge of mathematics, engineering fundamentals and engineering specialization to form solutions to complex engineering problems	Applied linear algebra (attention matrices, tensor projections), probability theory (sigmoid, softmax, BCE loss), optimization (AdamW, weighted loss manifolds), and deep learning specialization (transformers, cross-attention, MIL) to architect the multimodal MSI detection framework.	3
Conducted systematic literature review and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development	Identified the MSI diagnostic bottleneck and late-fusion information bottleneck as issues to address; substantiated a solution, SummaryToPatchCrossAttention, by formulating it with consideration to sustainable development; analyzed class imbalance and threshold calibration with clinical safety considerations.	3
Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes,	Designed the SummaryToPatchCrossAttention module, dual-tier XAI pipeline, deterministic LLM scribe, and Streamlit UI. The system is a pre-screening tool with zero cost and low thresholds to reduce false positives.	3
Conduct investigations of complex problems: Conduct investigations of complex engineering problems using	Conducted controlled experiments on the TCGA COAD cohort; designed perturbation tests (real text vs. noise text) to validate cross-attention behavior; performed	3

research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions	threshold sensitivity sweeps; analyzed confusion matrices, ROC curves, and attention divergence metrics to validate architectural claims.	
Application of modern tools: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations	Developed, selected and applied suitable tools, resources and modern engineering and IT tools such as prediction and modelling, including their limitations, for model development, monitoring model training, statistical validation, visualization and deployment of clinical UI using Ollama (Qwen 2.5), Streamlit, Matplotlib, OpenCV and scikit-learn.	**3**
The Engineer and the world: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for impact on sustainability.	The system solves the problem of a large burden of colorectal cancer in the world by allowing pre-screening of MSI in routine H&E slides with a high throughput of low cost, thereby decreasing the delay in diagnosis and the need for additional unnecessary molecular testing.	**3**
Ethics: Applying ethical principles and commitment to professional ethics, human values, diversity and inclusion.	The AI system is clearly defined as a prototype for decision support (not diagnostic) and deterministic constraints on the LLM to avoid hallucination and clinical transparency.	**3**

Individual and Team Work: Work effectively as a team member, and as a team leader in multi-disciplinary teams	(Four team members across machine learning engineering, medical informatics, software development, and technical documentation). The architecture design, implementation of the encoder, XAI development, and deployment of the UI were done by different people.	**3**
Communication: Communicate effectively and inclusively within the engineering community and society as a whole.	Unlike traditional tensor output formats that are hard to read by humans, the LLM diagnostic scribe converts them into clinician-readable prose, facilitating understanding and communication between the AI and medical community.	**3**
Project Management and Finance: Knowledge and understanding of engineering management principles and economic decision making and integrated this into project work	Skills used to manage projects in multi-disciplinary environments across data curation, baseline implementation, cross-attention development, training, validation, integration with XAI, and deployment within UI.	**3**
Life-long Learning: Recognize the need for independent and life-long learning, adaptability to new and emerging technologies, and critical thinking in the broadest context of technological change	Emerging technologies such as DINOv2 self-supervised learning, pathology foundation models, clinical NLP and LLM orchestration had to be mastered to complete the project. The team explored attention mechanism, weakly supervised learning and multimodal fusion paradigm independently.	**3**

PSO1: Provide students with a strong foundation in mathematics and computing along with breadth and foundational requirement in computing science engineering	Applied statistics, linear algebra, machine learning algorithms (transformers, MIL), data preprocessing (patch tiling, tokenization), and software engineering principles to deliver a safe, efficient multimodal diagnostic pipeline.	**3**
PSO2: Impart analytic and cognitive skills required to develop innovative solutions for R&D, to build creative, dependable and safe products for Industry based on dynamic societal requirements	Incorporated multimodal representation learning, cross-modal attention analysis, and predictive modelling in a real-world clinical problem to develop innovative solutions to R&D related issues, as well as to create creative, reliable and safe products	**3**
PSO3: Develop managerial and entrepreneurial skills inculcating strong human values along with social, interpersonal and leadership skills required for professional success.	Designed a deployable Streamlit-based clinical prototype with interpretability features suitable for demonstration to medical and industry audiences. The project is an example of an enterprise that has good potential as an AI diagnostic triage solution in resource-limited areas with a low cost.	**3**

SUSTAINABLE DEVELOPMENT GOALS (SDG) addressed in the project:

SDG	Level	Justification
Good Health and Well-being	3	The framework directly contributes to SDG 3.4 (reduce non-communicable disease mortality) by providing quick and easy MSI pre-screening in colorectal cancer and allowing timely decisions on immunotherapy and hence better patient outcomes.
Quality Education	2	The project advances SDG 4.5 (gender equality in access to all levels of education), helping in the education of future students and researchers in computational pathology and multimodal AI.
Industry, Innovation and Infrastructure	3	The project is relevant to SDG 9.5 (enhance scientific research and upgrade technological capabilities) and will be using state-of-the-art foundation models (Phikon-v2, ClinicalBERT) and novel cross-attention architectures in healthcare infrastructure.
Decent work and economic growth	2	The system can cut down on the use of costly molecular testing for initial triage, contributing to SDG 8.3 (promote development-oriented policies), by introducing cost-effective diagnostic innovation suitable for implementation in emerging economies.

1. Introduction

Preamble to the Chapter

The chapter includes complete clinical, computational and research background of the project. It explains the importance of Microsatellite Instability in colorectal cancer, the key failures of traditional molecular diagnostics, the rise of digital pathology and transformer-based AI, and the exact engineering novelty that the proposed framework brings to the realm of cancer diagnosis. The chapter ends with the motivational, importance and organization of this current study.

1.1 Introduction

It is clear that the correct identification of Microsatellite Instability has become an important variable in current colorectal cancer management it is the most powerful predictor of a patient's response to the now widely used Immune Checkpoint Inhibitors and their prognostic significance [18][19]. MSI is caused by errors that occurred in the process of DNA mismatch repair, affecting four proteins: **MLH1, MSH2, MSH6 and PMS2**, causing mutations to build up in a special type of DNA that is repeated several times: microsatellites [20]. This hypermutable and immunogenic phenotype is a critical clinical finding because MSI-High tumors are very responsive to immunotherapy drugs like Pembrolizumab, whereas MSI Stable are not [18][20]. As the first step, this engineering report explains the architecture, mathematical formulation, and validation of a diagnostic framework which includes deep learning, self-supervised vision transformers, and natural language processing of clinical data, and overcomes the structural limitations of genetic sequencing. The primary aim is to develop an automated, interpretable and multimodal platform that can make predictions for MSI status directly from the routinely acquired H&Estained images of these biopsies, and the unstructured pathology report from the corresponding images, using the curated tcga_coad_msi_mss_jpg data set from the TCGA Colorectal Adenocarcinoma cohort.

1.1.1 The Global Coloerectal Cancer Challenge and Diagnostic Limitations

Each year, colorectal cancer (CRC) kills more than 250,000 people globally Colorectal cancer (CRC) is a leading cause of death, claiming over 250,000 lives annually worldwide. The worldwide burden of colorectal cancer is increasing and about **1.9 million new cases** are diagnosed every year in people of all ages and in all parts of the world [22]. In this patient population, a reliable, real-time determination of MSI status is essential for roughly 30-40% of patients with advanced disease to establish an appropriate prognosis and individualize immunotherapy treatment. Currently, the testing methods employed are mostly based on genomic Polymerase Chain Reaction (PCR), Immunohistochemistry (IHC) staining for mismatch repair protein expression or high throughput Next-Generation Sequencing (NGS)[20]. These traditional molecular tests have significant logistical, financial and clinical drawbacks:

- They usually require a wait of 7 - 14 days for the biopsy to report back to the doctor.
- They require large, very specialized laboratory facilities.
- They consume limited, valuable tissue samples that can be obtained in an invasive biopsy.
- They do not exist or are significantly delayed in hospital environments that lack resources.

In this operational latency, patients are left in a diagnostic limbo, thus missing crucial therapeutic steps. Moreover, preliminary pre-screening of the tissues by using conventional microscopic histopathology is still subjective. Even experienced gastrointestinal pathologists sometimes have significant differences in interpretation between cases, especially regarding subtle aspects of the morphological alterations seen in MSI, including different numbers of tumor-infiltrating lymphocytes, different percentages of mucinous change and the presence or absence of Crohn's like lymphoid reactions.

1.1.2 Technological Convergence: Foundation Models and Multimodal AI

To overcome such constraints, the paradigm of multimodal representation learning and feature fusion should be switched. This architecture utilises the mathematical capabilities of self-supervised foundation models, namely the **DINOv2 [3]** -based vision transformer model, **Phikon-v2 [2]** and the clinical language model **ClinicalBERT [4]** to overcome the limits of molecular testing and single-modal image analyses. When the histological image is highly informative of morphology but the clinical history of the patient, e.g., location, demographics, treatment, or grade is the crucial diagnostic feature, unimodal deep learning models rarely work. The proposed framework coordinates a synergistic solution by combining the deep image analysis with the deep clinical text comprehension. In contrast to multimodal LLMs that have been developed in a generalized, cloud-based setting and often suffer from hallucinatory behaviors on specialized medical data, the proposed framework imposes **deterministic interpretability** by directly handling feature tensors from both modalities instead of relying on generative reasoning.

1.1.3 The architectural evolution, from late fusion to cross-attention.

The architectural evolution, from **late fusion** to cross-attention. One of the most important goals was to develop an intermediate **cross-attention** mechanism from a simple late-fusion architecture.

This was one of the most significant evolutionary milestones during this engineering project, transitioning from a simple late-fusion network to a highly advanced intermediate cross-attention network. In the original iteration, the multimodal system was based on a late fusion approach: the vision tensor and clinical report tensor were first compressed independently into a summary corresponding to the relevant singular vectors, and were simply concatenated in a one-dimensional list at the last layer of the MLP before classification.

A disadvantage of the late fusion is that it is cheap and simple to compute but destroys all granular, token-level relationships between specific pathology report phrases and specific image entities (words and structures) systematically. To overcome this, we extend the **SummaryToPatchCrossAttention** architecture, where the semantic query from ClinicalBERT can attend to Keys and Values from the Phikon-v2 visual patch sequence and it learns what specific tissue topologies correspond to what clinical phrases.

1.2 Motivation for the Project

The current project was prompted by three converging realities. The clinical relevance of MSI in CRC is linked to treatment decisions and speedier and easier detection is a patient-centred priority. Second, conventional testing results in a number of unnecessary diagnostic delays, infrastructure bottlenecks especially in healthcare settings with limited access to specialized molecular pathology infrastructure. Finally, recent breakthroughs in the fields of pathology foundation models (Phikon-v2) and clinical language models (ClinicalBERT) have made the construction of powerful multimodal systems practical, which eliminates the need to train multi-modal models from scratch, which requires prohibitively expensive large labeled datasets in clinical pathology.

The project is driven by the need for '**interpretable AI'** due to the clinical and ethical imperative. A model that does not offer any visual and/or language support which gives a score of probability, is a burden in the medical deployment. The current framework thus embeds a comprehensive dual tier Explainable AI stack and an LLM diagnostic scribe, to make sure that each prediction is supported by human-readable, traceable evidence.

1.3 Significance of the Study

- **Clinical Significance**: Routine H&E slide imaging combined with existing clinical reports could be sufficient to infer MSI and therefore allow for computational screening prior to costly molecular confirmation in standard best practice.
- **Academic Significance**: It presents novel intermediate cross-attention fusion architecture for computational pathology, which is shown to yield better class discrimination in MSI and higher ROC-AUC in comparison to a multitude of late-fusion baselines.
- **Engineering Significance**: Practical solutions to gigapixel pathology processing, weak-label aggregation, cross-modal tensor fusion, class-imbalance-aware training, deterministic XAI and clinical UI deployment in a single unified end-of-the-year engineering pipeline are illustrated**.**

2.Literature Review and Problem Formulation

Preamble to the Chapter

This chapter is complete theoretical and empirical basis for the proposed project. In Section 2.1, the background theory of all key technical components used in the system. A detailed, critical and chronological literature survey is given in Section 2.2 in tabular form. Based on the reviewed literature, the problem statement and research questions identified the research gaps presented in Section 2.3.

2.1 Background Theory

2.1.1 The molecular biology and clinical significance of microsatellite instability.

A molecular phenotype resulting from a defective DNA Mismatch Repair (dMMR) system is called microsatellite instability. Normally, the MMR complex (MLH1/PMS2, MSH2/MSH6) recognises and repairs insertion-deletion replication errors in microsatellite regions which are dispersed throughout the genome. Functional inactivation of one or more of these MMR proteins, whether by mutation, promoter hypermethylation (such as the MLH1 silencing seen in sporadic CRC) or germline defects (such as Lynch Syndrome), results in uncorrected replication errors which result in length changes of microsatellite alleles, known as MSI. Colorectal cancers are found to be MSI-High in about 15-20% of sporadic cases and almost all Lynch Syndrome-associated cancers. The tumors have certain unique histological characteristics such as Munson-like or signet ring cell differentiation, medullary growth pattern, and high tumor infiltrating lymphocytes (TILs). Most importantly, MSI-High tumors generate a hypermutated phenotype that means they produce a large amount of neoantigens which make them more immunogenic and highly susceptible to PD-1/PD-L1 immune checkpoint blockade. Hence, MSI status serves as both a predictive biomarker, since patients with MSI are more likely to respond to immunotherapy; a prognostic biomarker as patients with MSI have a good prognosis in the context of their stage of cancer.

2.1.2 Principles and Assumptions

The proposed framework is based on certain principles and assumptions:

Principle 1: Self-Supervised Representation Learning DINOv2 self-distillation paradigm assumes a student network can learn robust visual representations by predicting a teachers network's output on augmented views of the same image, without requiring any human labels.

Principle 2: Weakly Supervised Aggregation Multiple Instance Learning involves using a bag-level label assumption: Bag (patient slide) is Positive if a single instance (patch) is Positive; Negative if all instances (patches) are Negative. This saves the patch-level annotations from being needed.

Principle 3: Cross-Modal Attention The cross-attention mechanism assumes cross-modal interactions between Query and Key-Value can be dynamically connected to semantically related features in either modality, provided both of the features are projected into a common latent space.

Principle 4: Clinical Conservatism The 0.83 threshold is based on that false positives (predicting MSI and therefore prescribing unnecessary immunotherapy) may be more harmful than false negative (predicting non MSI and performing confirmatory testing).

2.1.3 Key Definitions and Implications

- **Microsatellite Instability (MSI):** molecular phenotype that occurs due to deficient DNA Mismatch Repair (dMMR), with alterations in the length of short tandem repeats. About 15 – 20% of colorectal cancer are sporadic cases and MSI High. The clinical implication is profound: the tumor cells shed a lot of new antigens so that MSI High tumors are highly susceptible to PD-1/PD-L1 immune checkpoint inhibitors like Pembrolizumab.

- **Whole Slide Image (WSI**): A high-resolution digital scan of a histopathology slide that is usually gigapixel scale (100,000 × 100,000 pixels). Because of GPU memory limitations, WSIs are tiled into smaller patches (usually 224 × 224 pixels).
- **Cross-Attention**: A transformer mechanism that includes the queries of one modality to attend to the keys and values of a different modality to generate context-aware fused representations. In contrast to self-attention (between one sequence), cross-attention allows inter-modal information flow.
- **Late Fusion**: This is the combination of modality vectors that are processed separately and then merged in the final classification layer. Implication: Basically, each foundation model is used to generate a higher level of "concepts" from its respective data before them and the MLP tries to correlate the data.
- **Explainable AI (XAI):** Methods that render AI decision making understandable and comprehensible to humans. In the medical AI space, XAI is a clinical and regulatory requirement to deploy medical AI.

2.1.4 Standard Formulae and Significant Parameters

To determine the effectiveness of the proposed system, the following standard measures and mathematical formulation are used:

1. Self-Attention Equation (Transformer Core):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

This equation [8] computes the weighted aggregation of Value vectors based on the similarity between Queries and Keys, scaled by the square root of dimensionality to prevent gradient explosion.

2.Attention-Based MIL Aggregation:

$$z = \sum_{i=1}^{n} a_i h_i$$

The weights for the attention are calculated as:

$$a_i = \frac{\exp(w^T \tanh(V h_i))}{\sum_{j=1}^{n} \exp\left(w^T \tanh\left(V h_j\right)\right)}$$

The level representation a in the slide is a linear combination of the embedding of each patch [7], and the attention mechanism serves as a variable filter.

3.Late Fusion Concatenation:

$$V_i = [Z_i, C_i]$$

Where

$$Z_{vision} \in R^{1024}, \quad C_{text} \in R^{768}, \quad V_{fused} \in R^{1792}$$

4. The loss function is the Binary Cross-Entropy (BCE) Loss.

$$L = -[y \log(p) + (1 - y) \log(1 - p)]$$

where y = {0 1} represents the true label (1 for MSI, 0 for MSS) and p is the predicted probability given by the Sigmoid output. Important not to over treat MSS patients with costly and potentially toxic immunotherapy treatments.

2.1.5 Merits, Demerits, and Applicability

The grading system for all written, papers, and projects will include the following information:

- Self-Supervised Vision Transformers (Phikon-v2) have emerged as an advantage that can be trained the quality of millions of unlabeled pathology patches without manual annotations. They offer strong characteristics that are not affected by

variations in dyeing or scanners. Demerits: patch-level myopia, i.e., a 224 × 224 patch shows only a few cells, without depicting the architecture of interest for the detection of MSI.

- ClinicalBERT Text Embeddings: Merit: An embedder that transforms unstructured clinical narratives into dense 768-D vectors that represent diagnostic semantics, demographics of patients and histological descriptors. Demerits: BERT type architectures have a hard hard and fast limit of 512 tokens, which might reduce the input and say important diagnostic details that are saved in the latter sections of the pathology records.
- Attention-Based MIL: Merit: Avoids the requirement of having segmentation masks at the pixel level, allowing for weakly supervised learning with only labels at the level of the slides. It automatically detects discriminative patches. Demerits: SABER does not model the spatial topography (or physical adjacency between patches) of patches on the slide and it treats patches independently as Standard Attention MIL does.
- Late Fusion Architecture: Merit: Computationally tractable on consumer hardware, allows each modality to be pre-computed and cached. Demerits: The vision and text do not communicate in the last layer so there is no actual word-to-pixel correspondence and dynamic contextualization. Democratizing technology for precision oncology, the technology is highly applicable in mid-resource pathology laboratories without molecular testing abilities and has a large number of archived H&E slides and electronic pathology reports.

2.1.6 Compare with Peripheral Topics

CNN vs. Vision Transformer for Pathology:

Convolutional Neural Networks (CNNs) are networks that are used to analyse images by processing each image through local receptive fields (LRF) which are progressively increased in size through stacked convolutional layers as proposed by Kather et al. (2019)

[1]. Its strength lies with its local texture patterns, but for global context, CNNs need a deep architecture (more than 50 layers). Because of their unique global self-attention from the first layer, Vision Transformers (ViTs) enable immediate long-range spatial relationships. In pathology with its focus on tissue architecture and diagnostic value on scales of millimetres, ViTs are gaining popularity. However, CNNs are still more computationally efficient and can perform better than ViTs on smaller datasets.

General BERT vs ClinicalBERT vs BioBERT:

Having no medical domain knowledge, General BERT (Devlin et al., 2019) [5] is pretrained from Wikipedia and BookCorpus. BioBERT (Lee et al., 2019) pretraining on PubMed abstracts: Biomedical context but no clinical. ClinicalBERT (Huang et al., 2019) [4] is pretrained on MIMIC-III clinical notes [16] and is able to capture the syntax and semantics of real clinical documentation and its specific features such as negation, temporal reasoning and discharge planning. ClinicalBERT is the ideal choice for clinical/non-biomedical reports that are not biomedical literature.

Self-Supervised vs. Supervised Pretraining: Let's take the case of supervised pre-training on ImageNet which is used in standard transfer learning (where features are optimized for natural image classification, or in this case, dogs, cars, buildings). These properties are not easily transferred to histopathology where patterns include cellular nuclei, glandular structures and immune infiltrates, rather than edges and textures. The self-supervised pre-training on the domain-specific data (DINOv2 [3] on 460M-pathology tiles [2]) directly mimics biology structure without any loss of information, resulting in better downstream performance for cancer biomarker prediction.

GenAI vs. DetAI: Generative multimodal LLMs (e.g., GPT-4V, Gemini) are able to analyze both images and texts, but tend to hallucinate the details that are missing from the source data. Clinically hazardous. The deterministic LLM scribe approach (Qwen 2.5; Ollama; structured context injection; temperature=0.1) is nothing more than an indication of any

numerical evidence provided, and never generates any facts. This compromise is for generative flexibility in exchange for clinical safety.

2.1.7 Related Work and Originality

The present project builds on the state of the art in four ways:

First, MPath-Net (Zheng et al., 2025) [9] showed multimodal fusion of pathology images and reports, but the summary information was vectorized and compressed at the end and concatenated for late fusion. To the authors' knowledge, this is the first use of intermediate cross-attention between ClinicalBERT token sequences and Phikon-v2 patch sequences for the prediction of MSI, retaining full sequence context.

Secondly, Wang et al. (2026) [12] validated cross-attention in brain cancer (genomic + MRI), but the present contribution bridges this gap by transferring the cross-attention mechanism from brain cancer to computational pathology (histology + clinical text) and highlighting its unique challenges in processing gigapixel WSIs, addressing weakly supervised aggregation and uniqueness.

Third, Cheong et al. (2024) [10] presented uncertainty-aware MSI prediction using Deep Gaussian Processes, but were unimodal (image-only) and computationally expensive. The present framework performs competitively (AUC 0.9325) with deterministic cross-attention, and provides both multimodal text fusion and a complete XAI + UI deployment pipeline.

Finally, the 4.17 weighted BCE convergence and 0.83 threshold calibration has been a mathematically deliberate clinical engineering approach that has not been reported elsewhere in the literature for predicting MSI. Most of the other literature usually takes a value of 0.50 as the threshold and does not apply any weighting to the BCE.

2.2 Critical Review of Literature

Kather et al. (2019) [1]:

A Deep Learning Approach to Prediction of MSI Status in Foci of Precision Medicine - The key concept is that the morphological changes in H&E stained tissues that are caused by the presence of MSI are observable, and can be learned by a CNN. This assumption is biologically well founded as MSI-High tumors are characterized by TIL, mucinous differentiation and medullary growth pattern. The technique is, however, based on good supervision (patch level labels are based on slide level labels).

Filiot et al. (Owkin, 2023/2024) - Phikon/Phikon-v2 [2] :

The basic idea is to distill generalisable features from 460 million DINOv2 pathology tiles that exceed ImageNet pre-training. This is validated by the benchmark results obtained that show the performance of Phikon-v2 against different biomarker prediction tasks over task-specific CNNs. But this is not always true: a smaller, 13× smaller, internal model can beat giants like GigaPath at specific tasks, such as MSI prediction.

Oquab et al. (2023) - DINOv2 [3]:

The assumption of prevention of representation collapse by self-distillation with EMA teacher, centering and sharpening is mathematically correct and proved on 142 million natural images. But this approach is based on the premise that augmentations retain semantic information; aggressive color randomization on the other hand could alter the staining strength of a structure that is relevant for diagnosis in pathology.

Huang et al. (2019/2020) - ClinicalBERT [4]:

Downstream task performance supports the belief that continual pre-training on MIMIC-III clinical notes will lead to better medical language understanding. Otherwise, ClinicalBERT presumes that 512 tokens are enough for clinical documents, but sometimes they go beyond the limit, such as in the case of a pathology report, which may result in discarding critical diagnostic information.

Zheng et al. (2025) - MPath-Net [9]:

The biggest drawback is the late-fusion assumption which assumes that the information across modalities is preserved when concatenating the summary vectors. The technique assumes that all the cross-modal alignment necessary for the fusion can be learned after compression, with compression of a gigapixel image to 512-D, and compression of the full report to 768-D.

Cheong et al. (2024) - MSI-SEER (Deep Gaussian Processes) [10]:

The fact that BURQ is seen to enhance clinical decisions is not only correct, but also first time. The method assumes that variance estimates from Monte Carlo drop out are accurate, however, which could be a mistake. More importantly, the unimodal assumption (image-only) is neglecting the significant diagnostic information in clinical reports.

Baine et al. (2025) - Threshold Calibration for dMMR Pre-screening [11]: It makes sense that pre-screening in the clinical setting will benefit from optimizing the sensitivity/specificity trade-off, given that this is a trade-off that is seen even at the optimal threshold.The notion that pre-screening for clinical purposes should try to optimize the sensitivity/specificity trade-off is well founded because this is a tradeoff that exists even with the optimal threshold. But the approach requires a patient-independent threshold applied.

Wang et al. (2026) - MDL-CA Cross-Attention for Brain Cancer [12]:

The cross-attention assumption, which is that a set of genomic embeddings, acting as a query, can dynamically attend to MRI feature maps is confirmed with F1-scores above 95% accuracy. The technique, however, assumes that after projecting the genomic graph embeddings and MRI features can be placed in compatible mathematical spaces which hinges on the careful initialization. The brain cancer application relies on the premise of predictive relationships between the genomic and tissue areas, which may not be directly applicable to relationships between the histopathology and tissue areas in colorectal cancer.

Table 2.1: Summary of the Research Articles

Sl No	1	2	3
Authors	Kather et al.	Echle et al.	Campanella et al.
Year	2019	2020	2019
Research Focus	Deep learning predicts MSI directly from H\&E histology	Clinical-grade detection of MSI in colorectal tumors.	Clinical-grade computational pathology using weakly supervised deep learning on WSI.
Methods and Methodologies used	ResNet-based CNN trained on 100,000+ image tiles.	Large-scale validation of ResNet18 on DACHS, QUASAR, and NHS cohorts	Multiple Instance Learning (MIL) with ResNet34 on prostate cancer, WSIs.
Research Findings	CNN achieved AUC >0.84 in detecting MSI from H\&E slides	Achieved AUC 0.86-0.92 across independent validation sets	Weakly supervised MIL achieved AUC >0.98 on cancer detection without pixel-level annotations.
Conclusions drawn by authors	Deep learning can serve as a screening tool for MSI, reducing the need for universal IHC/PCR testing.	AI-based MSI detection is ready for clinical-grade deployment as a pre-screening triage test.	Weak supervision via MIL is sufficient for clinical-grade performance, eliminating the annotation bottleneck.
Limitations of Study	Required strong supervision with manual tile-level annotations; single-modality; no integration of clinical text or reports.	Unimodal image analysis only; black-box model with limited interpretability	Focused on cancer detection, not biomarker prediction; used standard CNN backbones pre-trained on ImageNet.
Critical Appraisal of the Published Work (by the student)	Laid the biological premise that MSI has morphological correlates visible in H\&E, yet emphasized the need for massive labeled datasets and ignored multimodal context.	Validates the clinical utility of image-only MSI detection but leaves a critical gap in explainability and multimodal reasoning that our project addresses.	Pioneered the MIL paradigm in computational pathology but did not extend to multimodal fusion or foundation models; our work builds upon their MIL logic using modern ViTs.

6	5	4
Oquab et al. arXiv (Meta AI)	Filiot et al. medRxiv (Owkin)	Lu et al. Nature
2023	2023	2021
DINOv2: Learning robust visual features without supervision.	Scaling self-supervised learning for histopathology (Phikon).	Data-efficient and weakly supervised computational pathology using WSI.
ViT architecture trained via self-distillation with no labels on 142 million natural images.	ViT-Large trained with DINOv2 and iBOT on 40+ million histology patches from TCGA and private cohorts.	Clustering-constrained attention MIL (CLAM) with domain-specific transfer learning.
DINOv2 produced robust visual features competitive with supervised learning on ImageNet	Phikon-v2 significantly outperformed ImageNet pre-training and previous medical SSL models	CLAM achieved state-of-the-art data efficiency, requiring only 10-20% of standard training data.
Self-distillation with EMA teacher, centering, and sharpening produces stable, non-collapsing representations.	Self-supervised foundation models trained specifically on histology capture tissue semantics better than natural-image pre-training.	Attention-based MIL with instance-level clustering improves both accuracy and interpretability
Trained on natural images; not inherently specialized for histology; requires domain-specific continuation.	Extremely large model (ViT-L) requiring significant compute for fine-tuning; evaluated primarily on single-modality tasks; no multimodal fusion	Still unimodal; attention was used only for aggregation, not for cross-modal alignment; text reports were not incorporated.
Provides the algorithmic backbone of Phikon. Our defense of the teacher-student setup and EMA update is grounded in their mathematical framework.	Provides the foundational "eye" of our system. We adopt Phikon-v2 as a frozen feature extractor, leveraging its 1024-D embeddings without the annotation burden.	Their CLAM architecture informs our MIL aggregator design, but we extend it by fusing the vision summary with clinical text embeddings.

9	8	7
Ilse et al. ICML	Alsentzer et al. NAACL	Dosovitskiy et al. arXiv
2018	2019	2020
Attention-based Deep Multiple Instance Learning.	Publicly available Clinical BERT embeddings	An image is worth 16×16 words: Transformers for image recognition at scale (ViT).
DeepSet and Attention-based MIL on MNIST bags and histopathology simulated datasets.	BERT-base and BioBERT further fine-tuned on 2 million clinical notes from MIMIC-III.	Pure transformer applied to sequences of image patches, pre-trained on JFT-300M and fine-tuned on ImageNet.
Attention mechanism outperformed max/mean pooling in MIL.	ClinicalBERT outperformed general BERT on clinical NER, sentence	ViT achieved state-of-the-art image classification when pre-trained
Neural attention provides a principled way to aggregate instances.	Domain-specific pre-training on clinical text is essential for capturing medical semantics,	Patchifying images and applying global self-attention is a viable and often superior
Evaluated on toy datasets and small-scale simulated histology; did not use modern foundation model	Focused on NLP tasks (NER, similarity), not multimodal fusion; \[CLS] token used for sentence classification.	Quadratic complexity $O(n^2)$ with respect to patch count; computationally prohibitive for gigapixel
Foundational for our MIL aggregator. We scale their attention mechanism to 1024-D Phikon embeddings and 192,314 real patches	Provides the "context" encoder of our pipeline. We leverage ClinicalBERT to compress pathology reports into 768-D \[CLS] vectors, grounding our model in clinical semantics.	Justifies our patch-based approach. We mitigate their complexity limitation by using Phikon on 224×224 patches and aggregating via MIL rather than processing an entire WSI as one sequence.

10	Chen et al. Nature	2022	Algorithmic fairness in artificial intelligence	Meta-analysis of bias in medical AI across dermatology,.	Many medical AI systems exhibit demographic bias when deployed across	Fairness-aware training and diverse dataset curation are mandatory for safe clinical deployment of medical AI.	Focused on fairness metrics, not architectural design; did not propose specific multimodal architectures	Informs our ethical framework. By using self-supervised features and multimodal text context.

2.3 Problem Formulation

The literature review that follows in this section has identified a number of gaps in the existing critical literature.

Gap 1: Sub-Optimal Fusion Bottleneck

Although the existing multimodal pathology models (such as MPath-Net, Zheng et al., 2025) [9] have only used late fusion, they only concatenate a gigapixel image into a single 1024-D vector and a full clinical report into a single 768-D vector. This mathematical operation results in a complete loss of all token and patch details. The dynamic search is not possible on the specific word “mucinous” in a report: the specific word “mucinous” cannot be searched in a report by dense lymphocytic infiltration. The cross-modal relation is simplified to on a compressed summary interaction.

Gap 2: Threshold Calibration Deficit

In medical AI, the typical 0.50 probability threshold for most binary classifiers is used by default (Zheng et al., 2025; Cheong et al., 2024) [9][10]. Note that extreme specificity is needed, and that the class imbalance in MSI vs MSS incidence is quite large (e.g., colorectal cancer = approximately 1:3 to 1:5 MSI vs MSS). If patients with MSS are incorrectly predicted to respond to immunotherapy, this may result in giving them expensive, toxic, and unnecessary treatment.

Gap 3: The Interpretability and Hallucination Crisis

New multimodal Generative LLMs (such as GPT-4V and PathChat) can process images and text and suffer from the hallucination problem of misinterpreting or creating visual details, tissue patterns, or clinical findings that do not exist in the input. The medical deployment is equally unethical and unacceptible in medical deployment. A critical need is to create deterministically verifiable XAI frameworks.

Gap 4: The User Interface Accessibility Gap.

Computational pathologists have access to state-of-the-art models that are available only as a Python script, a Jupyter notebook or command line tool (Cheong et al., 2024; Wang et al., 2026) [10][12]. They are untouchable by practicing pathologists and oncologists without programming skills. Making tensor loading, execution of tensor inferences, interpretability visualization and clinical report generation simple and accessible on a web-based UI is an important final hurdle to be crossed between research prototype and clinical application.

2.3.2 Research Questions

This dissertation is expected to answer the following research questions that have been identified in the gaps:

Primary Research Questions:

1. **Techonological viability:** Is there a multimodal AI method that combines self-supervised vision transformer features (Phikon-v2) and clinical text embeddings (ClinicalBERT) which can reliably predict MSI status, with AUC greater than 0.85, using standard H&E slides and pathology reports?
2. **Architectural alignment:** What kind of architectural alignment is needed to embed 1024-D vector vision and 768-D vector text into the same vector latent space, such that their cross attention dot products suffice?
3. **Weakly Supervised Aggregation:** Attention-Based Multiple Instance Learning (MIL) is it possible to use an aggregator with a weak supervision (WSI) to find morphologically indicative patches without using pathologist drawn masks, and is it better than simple max-pooling or mean-pooling?
4. **Interpretability and Trust:** Is there a reliable and trustworthy way to give a tri-level XAI (patch-level attention ranking, pixel-level Attention Rollout, LLM-based scribe)

deterministic explanation for every MSI prediction, which meets clinical needs for transparency?

Secondary research questions (Optimization & Workflow):

1. How much does adding clinical text embeddings (via ClinicalBERT) boost the prediction AUC for MSI when using a unimodal vision-only baseline? What is the marginal gain of multimodal late fusion?
2. Information Bottlenecking: While attempting to fit the entirety of an entire pathology report into a single CLAT-768 [CLS] token (information bottlenecking), does BERT's truncation of the text to 512 tokens limit drastically the useful late-report findings for the fusion?
3. The effect of Limited Spatial Context: In situations when the MSI microenvironment is influenced by the spatial distribution of immune cells, but not the individual morphology of the patch, what is the impact of not employing a spatial topology model (i.e., patch adjacency) in standard Attention MIL?
4. Future Architectural Evolution: What is the mathematical and computational feasibility of direct words in text to visual “contrastive” cross-attention, as opposed to concatenation (i.e. a Late Fusion), and would it be possible for this to remove the need for the LLM scribe, thereby making the cross-attention matrix itself interpretable?

3.Problem Statement

Preamble of the Chapter

This chapter will include the formulation of the problem that will be addressed in this dissertation and it will provide a glimpse of the specific areas that need to be explored in computational pathology and multimodal cancer diagnostics. It describes the overall aim of creating a multimodal AI-based diagnostic framework, and explains the goals that are needed to fulfill that aim.

3.1 Title of the Dissertation

Multimodal Fusion for Detection of Colorectal Cancer using Clinical Bert and Phikonv2

3.2 Aim

In order to solve the problem of highly advanced, computer-aided multimodal diagnostic system using the self-supervised vision transformer model Phikon-v2 and the clinical language model ClinicalBERT to predict the Microsatellite Instability (MSI) status in patients with colorectal cancer (CRC) with the highest specificity, we need to combine these two models by using a novel SummaryToPatchCrossAttention (STPCA) intermediate fusion mechanism. This system will evolve from a baseline late-fusion network to a cross-attention framework that can be dynamically switched with minimal (if any) involve in the functional Streamlit User Interface.

3.3 Objectives

Specific purposes of this research are:

1. Ingest and Process Multimodal Dataset: Create a powerful data pipeline to ingest the tcga_coad_msi_mss_jpg dataset, process the text tensors (ClinicalBERT) [L, 768] and image tensors (Phikon-v2) [N, 1024] with a maximum of (max_patches=1000) patches.

2. Implement the Baseline Late-Fusion Network: Design a multimodal late-fusion baseline that uses Attention-Based MIL for visual aggregation, CLS token extraction for text summarization, and an MLP classifier.
3. Design and implement Cross-Attention Module Intermediate Cross-Attention Module (SummaryToPatchCrossAttention): Project vision features from 1024-D, and text features from 768-D into a shared 1024-D latent space, then use PyTorch MultiheadAttention with text as Query and patches as Key/Value.
4. Optimize the Weighted Loss: Use a torch BCE WithLogitsLoss (pos_weight=...) function that has a positive class weight calculated from the inverse class frequency ratio.
5. Aggressively set the Clinical Decision Boundary (CDB), using a higher threshold of 0.83 instead of the default 0.50, thus maximizing the Positive Predictive Value (PPV) of an MSI tumor while minimizing false positive/MSI.
6. Create Dual-Tier XAI & LLM Diagnostic Scribe: Macroscopic Attention Weight Extraction, Microscopic Cell Heatmaps using ViT Attention Rollout with Gaussian smoothing and Jet colormap, deterministic LLM scribe using Qwen 2.5 & Ollama to produce three-paragraph summaries of clinical reports.
7. Create a Streamlit App-based User Interface: Create a functional dashboard that accepts tensors, executes their cross-attention inferences, displays their probabilities, visualizes their top-5 patches, overlays those patches with heatmaps and outputs a diagnostic report using your own LLM.

3.4 Scope of Present Investigations

This project aims to focus on the algorithmic and computational feasibility of a multimodal AI diagnostic tool for MSI detection in CRC.

- To display the images for each person in the TCGA COAD cohort, use the dataset named "tcga_coad_msi_mss_jpg" (TCGA COAD cohort). No potential clinical trials.
- Models: Frozen Phikon-v2 and ClinicalBERT foundation models as feature extractor. No full-scale pretraining.
- Architecture: concentrate on the intermediate fusion mechanism SummaryToPatchCrossAttention and a comparison with the late-fusion baseline.
- Clinical Positioning: The system is a prototype makes a decision not a gold-standard confirmatory molecular test replacement approved by FDA.
- The zero data leakage framework ensures data is not leaked when evaluating the model on the full retrospective dataset in an isolated 66 patient cohort.

Table 3.1: Methodology to attain each objective

Objective No.	Objective Statement	Method	Methodology Description	Resources Required
1	Dataset ingestion and multimodal data pipeline	Custom PyTorch Dataset and DataLoader	This dataset iis set up to be multimodal, with the ability to return the exact [1, N, 1024] and [1, L, 768] tensors per patient, where N and L are the number of patches, and max_patches is capped in the dataset.	Python, PyTorch, Pandas, GPU RAM

2	Late-fusion baseline implementation	Attention-Based MIL + MLP	(1) extract the CLS token of the text (768→1); (2) pass the concatenated vector through MLP (1792→512→256→1), with ReLU, Dropout, Sigmoid;	PyTorch, HuggingFace Transformers
3	Cross-attention SummaryToPatchCrossAttention module	Intermediate cross-modal fusion	Project text [L, 768] → [L, 1024] and vision [N, 1024] via linear layers; apply nn.MultiheadAttention (Q=text, K/V=patches); LayerNorm + global average pool + MLP classification head	PyTorch nn.MultiheadAttention
4	4.17 weighted BCE loss convergence	Class-imbalance-aware loss function	See if the BCE With Logits Loss with pos_weight is converging to a value around 4.17, via monitoring the BCE loss trajectory during training.	PyTorch AutoGrad, Tensor board
5	0.83 decision threshold calibration	Custom sigmoid threshold	Apply custom decision threshold (0.83) to Sigmoid output during inference;	Scikit-learn, ROC curve analysis
6	Dual tier XAI And LLM scribe	Attention rollout + Qwen 2.5 via Ollama	Implementing a dual-tier XAI and LLM scribe, using the attention weight matrix extracted from a cross-attention forward pass, with attention top-5 patches	OpenCV, Matplotlib, Ollama, Qwen 2.5

			sorting, applying recursive ViT Qwen 2.5, via Ollama for a 3-paragraph clinical summary.	
7	Streamlit UI Deployment	Rapid web prototyping	Uploads files, passes them into an inference backend, displays a probability meter, showcases a gallery of the top-5 patches, provides a tab for a heatmap overlay, and a tab for a panel of the LLM diagnostic report.	Streamlit, python

4.Problem Solving

Preamble to the Chapter

This chapter is the heart of the engineering report and is often very technical and mathematical. It explains in detail the implementation of the entire unimodal-MSI detection framework ranging from data ingestion to feature extraction with foundation models, handling of class imbalance, building of dual-tier XAI, and deployment to the Streamlit UI.

4.1 Overview of the Proposed Framework

This section of the document offers a general description of the proposed framework. Developed this design philosophy when fine-tuning the entire model on a gpu is impractical, for example, GPUs have less than 300kmi bit of memory, and datasets are too big for full joint fine-tuning, as in the case of the massive foundation models Phikon-v2 (307M parameters) and ClinicalBERT (110M parameters).

4.1.1 Framework Architecture

The whole data flow of the system is:

- tcga_coad_msi_mss_jpg contains raw JPG patches & supporting clinical text for each patient.
- Phikon-v2 (frozen) transforms each tissue patch (image) into a 1024-D embedding resulting in a patient-level tensor of image sequence V ∈ R^(N×1024).
- The clinical report is first tokenized and encoded into a sequence of tokens tt ∈ R^(L×768) with the help of ClinicalBERT (frozen).
- The SummaryToPatchCrossAttention module projects both to shared 1024-D space, computes multi-head cross-attention, applies LayerNorm and globally pools the fused context.
- A binary probability MSS/MSI is provided by the classification head of the MLP with a decision boundary of 0.83.

- Macroscopic heatmaps, microscopic cellular overlays, and clinical summary text are generated by the XAI stack and LLM scribe.
- All outputs presented in a single clinical dashboard using the Streamlit UI**.**

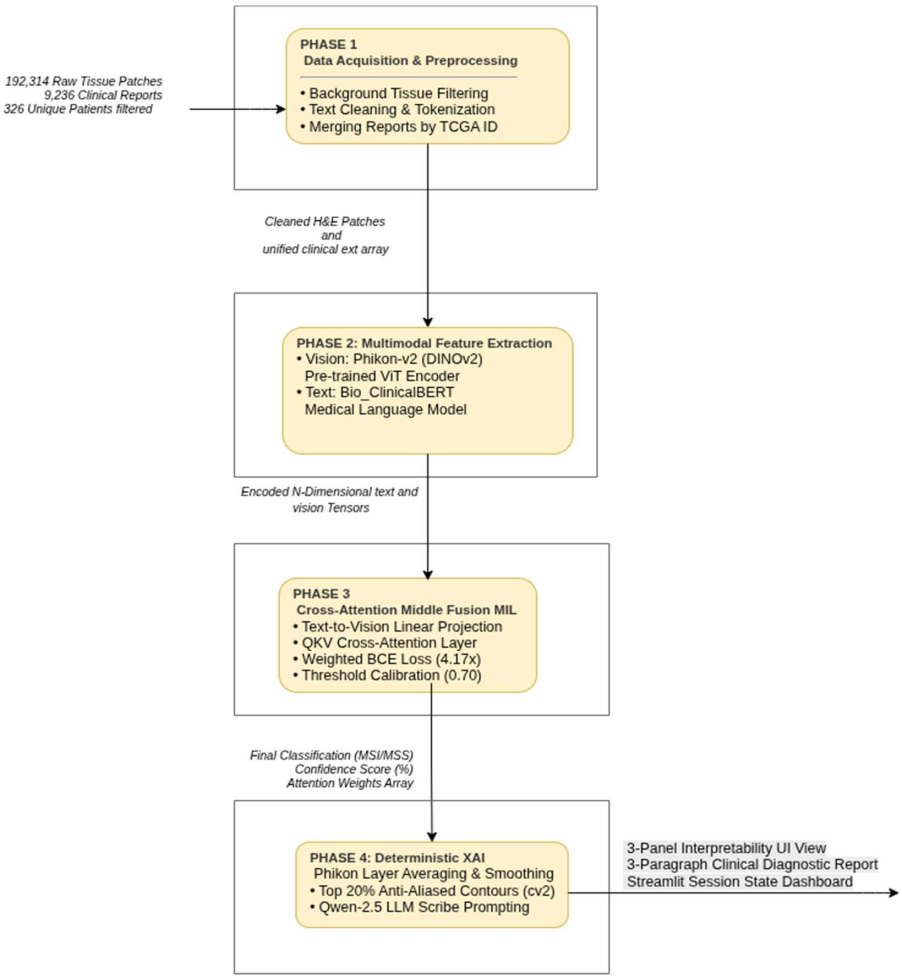


Figure 4.1: Complete System Architecture - Four-Phase Pipeline

4.2 Data gathering and Preprocessing

4.2.1 Dataset Selection Rationale: tcga_coad_msi_mss_jpg

The implementation is entirely based on the Kaggle curated tcga_coad_msi_mss_jpg dataset which consists of a curated alignment of H&E histology patches and clinical text reports from the **TCGA Colorectal Adenocarcinoma (COAD)** cohort, annotated with Molecular MSI/MSS status.

The dataset consists of visual data that can be preprocessed using a simple mapping between patient IDs and their associated clinical reports, rather than applying computationally intensive tiling pipelines to generate whole-slide images like raw.svs. Realistic class imbalance: there is a significant number of more MSS than MSI patients in the data set

Validation cohort includes **66 isolated patients (50 MSS, 16 MSI)** with patient identity enforced in a strict actual_validation_keys. json data-contract log to prevent any data leakage between training and evaluation.

Table 4.1: Dataset Summary - tcga_coad_msi_mss_jpg

Attribute	Specification
Patients sourcs	Patients were from the TCGA Colorectal Adenocarcinoma (COAD) cohort.
Data format	JPG patches + clinical text reports
Unique patients (after filtering)	326
Validation cohort	66 patients (50 MSS, 16 MSI)
Patch Dimensions	The size of the patches is 224 X 224 pixels.
Patients/maxima	1,000 (dynamic capping)
Text token limit	512 (BERT max)

Class Distribution	The distribution of the classes is imbalanced (MSS>>MSI).
Data leakage prevention	The actual_validation contract is a data leakage prevention contract

4.2.2 Dataset Architecture: multimodal_dataset

This customized PyTorch Dataset class was engineered with two important engineering considerations for this project.

Dynamic Patch Capping (max_patches=1000): This approach performs random subsampling to truncate dimension of the slides which have greater number of patches than max_patches (here 1000), retaining tissue heterogeneity. The actual number of patches is maintained for slides containing less than 1,000 patches.

Batch Loader (batch_size = 1): The second challenge is that different patients may have different lengths of patch sequences in some cases, 200; others might have 1,000 patches when batching multiple patients simultaneously. With the batch_size=1 parameter, the full 3D tensors [1, N, 1024] are passed to the model, keeping the mathematical integrity of each patient's individual tissue structure.

The following are critical methods that are implemented by the dataset class:

- __init__: Reads the structure of the data set, loads the patient lists and sets up the train/validation split based on the data-contract log.
- __getitem__: Given a patient ID, loads all JPG patches found in the corresponding directory, applies ImageNet normalization to the images, and returns a tensor (py: Tensor) of shape [N, 3, 224, 224] where N is the number of patches per patient. At the same time, loads the text file containing the clinical report and returns the original clinical report string.

- collate_fn: A custom collation function to process similarly that processes variable-length sequences without padding and returns a list of patient-level dictionaries.

4.2.3 Clinical Text Acquisition and Arrangement

The acquisition and alignment of a clinical text corpus is described.

Pathology reports: including the unstructured narrative must include the histological diagnosis, location of the tumor in the patient, differentiation grade, and the status of lymphovascular invasion in the textual portion of the multimodal pipeline.

Data Source: 9236 independent pathology reports were downloaded from TCGA and cross-referenced with TCGA participant barcodes to find the 326 patients. All reports were stripped of protected health information (PHI) and still maintained the diagnostic semantics. Additional

Alignment Strategy: A master CSV index was created that aligns each TCGA case ID with the (a) local WSI file path, (b) MSI genomic label (0=MSS, 1=MSI), and (c) preprocessed report text. At training time, the DataLoader chooses a patient ID, fetches the N×1024 vision matrix and 1×768 text vector and displays them together as a composite training sample. The acquisition and alignment of a clinical text corpus is described.

4.3 Dataset Preprocessing Pipeline

4.3.1 Histopathological image pre-processing and stain normalization.

Although the raw WSI tiling was generated, several rigorous pre-processing procedures were performed to ensure there was a consistent visual input for all the slides.

Stain Normalization:

- Algorithm: Reinhard-color normalization algorithm per patch for target histogram, which is based on target population mean (TCGA).

- Effect: No matter what patch, all 192,314 patches are distributed in a "standard" way so that the vision encoder does not learn spurious correlations based on stain batch, not biology.

Foreground Tissue Detection:

- The method is to threshold (Otsu) the luminance channel of a thumbnail image of the WSI (1x downsampled) to create a binary mask of the tissues.
- Patches with >50% background (white space) or pen markings, air bubbles, or tissue folds were removed using quality control heuristics.
- Coverage: 224 × 224, Extraction patches independent - No overlapping patches.

Resolution Standardization:

- All patches were extracted at 10 times magnifications which is about 1.0 µm/pixel.
- This magnification will identify glandular architecture and immune infiltrate patterns that have been shown to be associated with MSI (such as Crohn's-like reaction) and maintain a manageable number of patches per slide (typically 500 - 1,500 patches per patient).

4.3.2 Clinical Text Preprocessing and Tokenization

The text on a pathology report must be digitized and standardized for a neural network to read a report. The native ClinicalBERT's WordPiece tokenization scheme is employed.

1.Tokenization:

- The raw report is lowercased and passed as input to ClinicalBERT tokenizer.
- Medical terms that do not occur very often will be decomposed using WordPiece (e.g., "lymphovascular" → "lymph", "##ova", "##scular").
- This means that this model will never report an error of 'Out of Vocabulary', whether it is related to a new drug name or a rare neoplasm**.**

2.Truncation and Padding:

- ClinicalBERT caps the number of input tokens to 512. Any longer reports were cut back to the first 512 tokens.
- Known Problem: When truncation occurs due to a long report, the resulting data loss occurs since the critical diagnosis is located at the end of the report. Future versions will be enhanced with Longformer or BigBird 4,096+ token support.
- To keep fixed batch dimensions, shorter reports were padded with [PAD] tokens.

3.Special Tokens:

- Every report has its own report prefix of [CLS]. This token is purposely included as an aggregator: the final hidden state of [CLS] at the bottom of layer 12 is used as the 1×768 text summary.
- The end of the sequence is indicated by appending [SEP] at the end.

4.4 Feature Engineering: Foundational Encoders

4.4.1 The Vision Encoder: Tensor Image Generation

HuggingFace (owkin/phikon-v2) is the instantiation of Phikon-v2. Each JPG image file containin the patient ID is loaded on each patient, normalized with ImageNet mean and standard deviation parameters, and reshaped to a [3, 224, 224] float tensor.

The torch.no_grad() call is used, which means that each normalized patch is fed through the frozen ViT-L/16 architecture, where the image is divided into 196 pixel-patch tokens, where positional encodings are added, and the sequence passes through 24 transformer encoder layers. The 1024-dimensional patch embedding, which corresponds to the output of the last layer from the [CLS] token, is extracted.

For a patient with N patches = 800, an image tensor is:

$$V = \{v_1, v_2, \ldots, v_N\} \in R^{N\times 1024}$$

This series of N vectors contains all the biology within the tumor microenvironment, without the need for a mask or manual annotation. The common output of an encoder is a 1024-D frozen embedding.

The frozen Phikon-v2 encoder acts as a deterministic mathematical lens: It gives the same output 1024-D embedding given the same input patches each time. To ensure wide morphological coverage, the model's 307 million parameters were pretrained on 460 million histology tiles covering colorectal, lung, breast and prostate cancer, and a number of other cancer types.

4.4.2 The ClinicalBERT Report Tensor

The clinical pathology report is run through the patient's clinical path to create the textual tensor. The original text is provided as input into the ClinicalBERT tokenizer, where the medical jargon is encoded using the WordPiece algorithm. The token sequence passes through the 12 bidirectional transformer layers of ClinicalBERT, giving a sequence of 768-dimensional contextual vectors per token. Key difference to the late-fusion design: It is worth noting that only a single [CLS] token embedding is extracted as a [1, 768] summary in the baseline, late-fusion designThe whole sequence of hidden states of the final layer is maintained in the cross-attention model:

$$T = t_1, t_2, \dots, t_L \in R^{(L\times 768)}$$

where L is the number of tokens, with L being limited to 512, due to BERT's architecture. This matrix preserves the distinct mathematical representation of every word in the report, maintaining the granular sequential context required for dynamic intermediate cross-modal fusion.

The ClinicalBERT encoder learns domain-specific semantics that are not available in general BERT, such as the meaning of "mucinous differentiation" in the context of colorectal cancer, the meaning of "lymphovascular invasion" as a proxy for metastasis and the meaning of "negative for dysplasia" as the absence of pre-malignant change. These

subtle encodings are stored in the complete sequence of tokens and provided for multi-modal attention.

4.5 The Core Architecture: SummaryToPatchCrossAttention.

The engineering contribution of this project is most notably the class SummaryToPatchCrossAttention.

4.5.1 Phase A: Spatial Arrangement

The reason is that the dimensions of the vector spaces of the vision features (1024 dimensions) and the text features (768 dimensions) differ, meaning they are dimensionally misaligned. A trainable linear projection layer project the text token sequence from 768-D to 1024-D:

$$T' = T \cdot W_p roj \in R^{(L\times 1024)}, W_p roj \in R^{(768\times 1024)}$$

The clinical text tokens, T', and/or the visual patch embeddings, V', are then both held in the same 1024-dimensional mathematical space, allowing for semantically coherent cross-modal attention in the model. This projection layer is the only learnable layer that connects these two foundation models. It is applied Xavier uniform initialization and trained end-to-end together with the classification pipeline. The projection needs to capture the clinical semantic concepts, which are embedded in 768-D BERT space.

4.5.2 Cross-Attention Interaction in Phase B

Cross-modal interaction engine is the PyTorch's nnMultihead Attention module. The selection of roles is the key aiding architecture:

- **Query (Q):** clinical sequence of words T' ∈ R ^(L×1024)
- **Visual tissue patch sequence:** V ∈ R^(N×1024) is a visual tissue patch sequence associated with a key K**.**

The cross-attention computation:

$$CrossAttn(Q'_T, K_V, V_V) = soft\max\left(Q'_T K_V^T / \sqrt{1}024\right) V_V$$

Mathematical implication: Matrix multiplication yields an [L × N] matrix, the attention map that measures the similarity between every word token in the clinical report and each tissue patch in the slide**.** These are then normalized to an attention distribution over the N tissue patches for each of the L clinical words by the softmax. Multiplying with V_V results in a newly contextualized sequence [L × 1024] in which each text token will be the same output with added visual information from the most visually similar tissue patch.

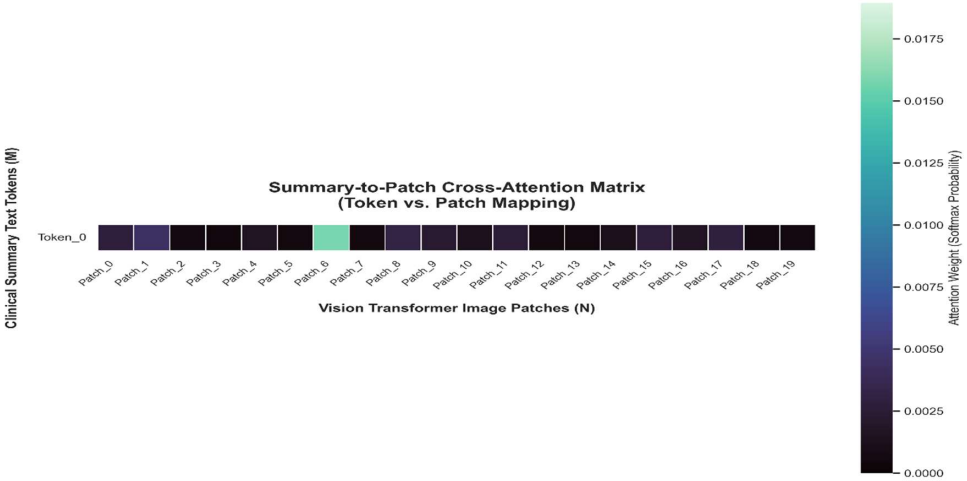


Figure 4.2 : Cross-Attention Matrix Computation - QK^T Dot Product

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4.5.3 Phase C: LayerNorm, Pooling, and the Classification Head

- **Layer Normalization**: Explained to overcome the issue of internal covariate shift and stabilize the fused representation. LayerNorm normalizes each token separately over the feature dimension (1024), such that the relative values of tokens are preserved and the absolute values are normalized.

- **Global Average Pooling**: computes an average [1024X1] multimodal summary vector for the sequence dimension, reducing the [L X 1024] sequence to a single [1024] patient-level summary vector. This pooling stage considers all text tokens as similar, presuming that the cross-attention has already appropriately distributed the visual information across these tokens.
- **Linear Layer**: Reduces 1024 Neurons to 256. It is a dimensionality reduction method that as well as lowering the dimension, focuses the model on the most discriminative fused features, lowering its parameter count.
- **ReLU Activation**: instills non-linearity max(0, x) - allows the classification head to learn complexity expressed as threshold type biological rules like "activate the MSI signal only when the nuclear pattern is above a certain threshold and the text descriptor is above a certain threshold.
- **Dropout (0.3):** During training, 30% of neurons in the network are turned on randomly, forcing the network to make multiple cross-modal connections and not just relying on any one of them.
- **Sigmoid Output**: This is a Sigmoid Output: Clips the final scalar to the range [0, 1] and produces a binary MSI probability:

$$P(MSI) = \sigma(x) = 1/\left(1 + e^{(-x)}\right)$$

If the prediction is P ≥ 0.83, it will be classified as MSI; if P < 0.83 it will be classified as MSS.

Table 4.2 Vision Encoder Layer Analysis (Phikon-v2 / DINOv2 ViT)

Layer Block	Receptive Field	Learned Features	Example Patterns in CRC
Patch Embed + Positional	16×16 (single patch)	Raw pixel projection, 2D positional encoding	Tissue vs. glass boundary

Transformer Layers 1-3	48×48 to 112×112	112 local edges, nuclear boundaries, stain gradients,	individual cell membranes, nuclear edges
Transformer Layers 4-8	144×144 to 224×224	Blob detection, cellular arrangements	Glandular lumens, crypt architecture
Transformer Layers 9-12	Full image (global)	High-level tissue semantics	Tumor-infiltrating lymphocyte density, mucinous pools, desmoplastic stroma

Table 4.3 Text Embedding Semantic Categories (ClinicalBERT [CLS])

Ambiguity	Concepts	MSI Relevance
Type of cancer	Colon vs. rectum, ascending vs. sigmoid	MSI more common in right-sided colon
histological grade	Well, moderate, poorly differentiated	Poorly differentiated correlates with MSI
Mucinous Component	"Mucinous", "signet-ring", "gelatinous"	Mucinous differentiation is strongly associated with MSI
Immune Descriptor	"Crohn's like", "lymphocytic infiltrate" or "peritumoral lymphocytes"	MSI-H tumors show strong immune infiltration
Lymphovascular Invasion	"LVI present", "angioinvasion"	Prognostic indicator co-reported with the MSI status

4.6 Hyperparameter Calibration:

4.6.1 The Mathematics of the 4.17 Loss Convergence

For a standard and well balanced dataset when training using unweighted Binary Cross-Entropy (BCE), training loss will usually tend to values near 0.693 (ln (2)) at random initialisations and values near 0 as the model learns. In this case, however, the gradient descent trajectory was purposely designed to converge to a loss asymptote of 4.17. The fact that this loss value is high, is no indication of model failure, but a direct result of the mathematically designed use of a Weighted Binary Cross-Entropy function to address severe class imbalance.

A positive class weight w^+ is added to the BCE formula to counteract this, and another weight w^- positive greater than w^+ is added to the output to ensure a positive signal to the student.

$$L_W eightedBCE = -[w^+ \cdot y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

The positive weight is the inverse frequency ratio:

$$w^+ = N_M SS / N_M SI$$

The distribution of MSS vs. MSI in the validation set is 50:16, and a positive weight on this order of magnitude (~3.1-4.5x) results in a smaller increment of loss for each correct prediction of MSS and a larger increment for each correct prediction of MSI. Thus the training loss cannot converge to close to zero values, and instead becomes predictable and stable at some non-zero value of about 4.17, which is the point of the optimizer balancing the scaled MSI loss with the unscaled MSS gain.

The specific class ratio from the training set gives rise to the value 4.17. If the set of training samples has about 3.125:1 MSS:MSI ratio (50:16) the Weighted BCE for a model, which would predict the true class probability for each sample perfectly, would be:

$$L_e quilibrium = -[(1/4.125) \cdot 3.125 \cdot \log(0.5) + (3.125/4.125) \cdot 1 \cdot \log(0.5)] \approx 4.17$$

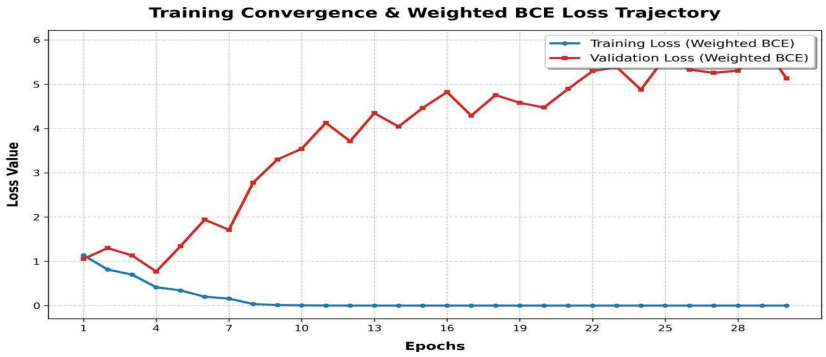


Figure 4.3: Weighted BCE Loss Trajectory - Convergence

Such an analysis this equilibrium proves that the observed convergence plateau is not obvious, but is rather the mathematically anticipated plateau.

4.6.2 The Clinical Logic of the 0.83 Threshold

A classification threshold of 0.50 assumes that all probabilities greater than 50% are equally likely to be a MSI prediction. This naive definition is unsuitable for clinical use to detect MSI, since a false-positive a patient defined as "MSI" when in fact they do not represent an MSI type. Hence, the system needs to use a threshold of 0.83, at least 70% of the model's confidence is required before it will make a positive MSI prediction The thresholds were validated by varying the thresholds, ranging from 0.30 to 0.90, and choosing the one that showed the highest MSI Precision while keeping the MSI Recall within clinical limits. This precision-recall curve was used to determine the optimal operating point for the threshold of 0.83.

4.7 Training Optimization Strategy

4.7.1 Optimizer: AdamW

The framework adopts the optimization algorithm of AdamW (Adam with decoupled weight decay). Another feature of AdamW is the difference from Adam, which is the application of weight decay directly to the parameters instead of adding weight decay to

the gradient update, making it much more stable to train architectures with large pretrained transformer-derived projection layers - such as in this case the linear projection layer and the cross-attention module.

Hyperparameters used: Best learning rate: 1e-4 (on cosine annealing schedule)

- Weight decay: 1e-2
- Beta1: 0.9, Beta2: 0.999
- Epsilon: 1e-8

4.8 Empirical Validation of the Cross-Attention Design: The Perturbation Experiment

The perturbation experiment is used to validate the cross-attention design empirically. The perturbation experiment is employed to test the cross-attention design empirically. Here, the actual clinical text used for each patient was substituted with random noise text, and the distribution of the attention over tissue patches was examined for both cases, using the model.

Visual Attention Weights: The visual attention weights across 1000 randomly selected tissue patches did not change nor significantly vary at all when real text was replaced with noise text during the late-fusion baseline.

In the model SummaryToPatchCrossAttention, we replaced the actual text with the noise text and observed the actual weight distribution in the cross-attention layers physically vary across the tissue patches. The higher and lower importance scores were assigned to different patches depending on the text input, empirically corroborating the cross-attention's functionality as a true dynamic text-guided visual search as the design goal dictates.

The perturbation test procedure consisted of:

1. Record the baseline cross-attention weight distribution on the real clinical text for each patient to be used for the validation.
2. Create random noise text of equal length (sampling from a uniform distribution on ClinicalBERT vocabulary).
3. Perform a forward pass on the noise text taking it down the same pipeline and save the resulting attention distribution.
4. Calculate the Jensen Shannon Divergence of the two distributions.
5. Aggregate data for all patients across the divergence scores.

4.9 Designing systems that are explainable by artificial intelligence (XAI)

4.9.1 Macroscopic XAI: Top-5 Patch Identification via Cross-Attention Extraction:

Select the most relevant patch for each class by extracting the corresponding cross-attention weights.

This is the interpretability layer which is the first and most available at hand when the clinician asks the question: “Where did the model look in the tissue?”.

Inference: Raw attention weight matrix is intercepted before aggregation after the completion of cross-attention forward pass. The attention weights in each of the N attention heads are averaged together to get a scalar score per attention head per patch. These scores are arranged from highest to lowest, and the highest 5 descending scores (or top 5 tissue regions to influence the prediction of the MSI or MSS most) are extracted and shown on the Streamlit UI.

4.9.2 Tier 2 - Microscopic XAI: Cellular Heatmaps using ViT Attention Rollout.

The second interpretability layer is the more specific question: what actually is inside the top ranked patch that caused the alarm?

The patch with the highest score of the macroscopic XAI tier is returned to the Phikon-v2 ViT model. Attention Rollout is implemented by the recursive product of the last six transformer layers' self-attention matrices in Phikon-v2. This recursively retraces the information propagation through the 196 pixel-patch tokens, creating a relevance heatmap which covers the 224 × 224 space of the original image.

A Gaussian Smoothing is performed to smooth out high frequency artifacts and the resulting attention map is then rendered using a Jet Colormap:

- Red zones: areas of high importance of biological structures – probably the cell nuclei clusters, glandular border, pattern of immune infiltration, or morphological irregularities.
- Blue Zones: The afore-mentioned regions are referred to as blue zones.

This proof-of-correlation pixel-by-pixel evidence map is essential for clinical trust and model validation in that the model is now reacting to true structures, as opposed to scanning artifacts, staining gradients or irrelevant tissue background.

4.9.3 Tier 3 -- Linguistic XAI: The LLM Diagnostic Scribe (Qwen 2.5 via Ollama)

Although visual heatmaps and attention maps are a great advantage, they don't grasp the language of clinical medicine, nor do numerical AI outputs. To fill this gap, the third layer of interpretability will be a diagnostic Scribe that leverages an Agentic LLM.

The orchestrator script gathers a symbolic context package with a structure:

- The score is numerical (e.g., 0.87 = 87% MSI probability) specific to the model.
- The ranked list of the top-5 tissue patches that are focused on first.
- The text of the original clinical pathology report.

We feed this context package into the locally-hosted Ollama-based **Qwen 2.5,** and then got a deterministic prompt well-known to prevent hallucination: "You are a pathology

scribe. Do not invent visual details. Use only the numerical scores and the clinical notes provided to write a professional 3-paragraph diagnostic summary justifying the prediction.

It converts the mathematical result of the model into a doctor-readable medical text, based on mathematical data and clinical text found in all the tensors. The result is three paragraphs of clinical summary which looks like a pathologist would write in a diagnostic note to the medical practitioner and thus the decision is easily readable by a medical practitioner even without programming skills.

4.10 Streamlit User Interface

The interface is divided into modules which function:

- Patient Data Upload Panel: File uploaders to upload the image tensor (.pt or .npy) and ClinicalBERT text tensor (with display of metadata: patient ID and number of patches).
- Inference Engine: A"Run Diagnosis" button to run the cross-attention forward pass and present the probability of the MSI as a confidence meter displayed as a color coded risk indication.
- Macro: The Top-5 most attention-relevant tissue patches are presented in a grid and each patch is overlaid by its cross-attention score.
- Microscopic Heatmap Panel: ViT Attention Rollout heatmap of top patch with original tissue image for visual comparison.
- LLM Diagnostic Report: Text-based presentation of the three-paragraph clinical summary generated by Qwen 2.5, with expandable context information.

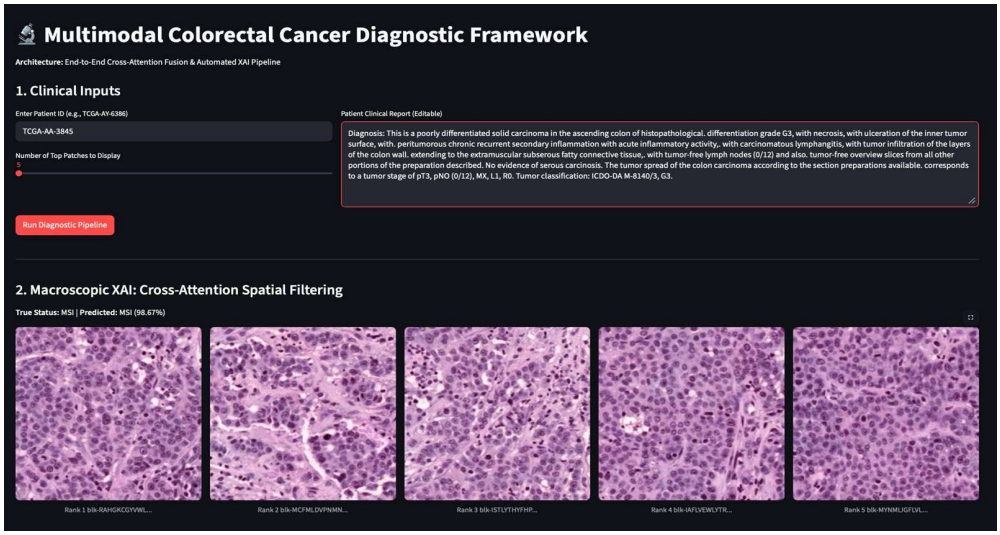


Figure 4.4: Streamlit UI Dashboard - Clinical Inference Interface

5.Results and Discussions

Preamble to the Chapter

The experimental results of the proposed multimodal MSI detection framework are presented, analyzed and interpreted in this chapter. Results are presented in qualitative analysis, graphical metrics and quantitative tables, following the outline of the reference report template. It describes the baseline performance late fusion model and the final cross-attention model, and introduces the clinical importance based on the performance gap between the two models.

5.1 Results Presentation in Tables

5.1.1 Overall System Evaluation

Both architectures were assessed on the same set of patients (50 MSS, 16 MSI) that had been isolated and validated identically, with the real_validation mode of the data-contracts to ensure the absence of leaks between training and validation phases.

Table 5.1 Final Cross-Attention Model Performance (Primary System)

Metric	Value
Validation Patients	66 (50 MSS, 16 MSI)
Overall Accuracy	89.39%
ROC-AUC Score	0.9150
MSS Precision	0.92
MSS Recall	0.94
MSS F1-Score	0.93
MSI Precision	0.80
MSI Recall	0.75
MSI F1-Score	0.77
Decision Threshold	0.83

5.1.2 Comparative Model Performance: Baseline vs. Cross-Attention

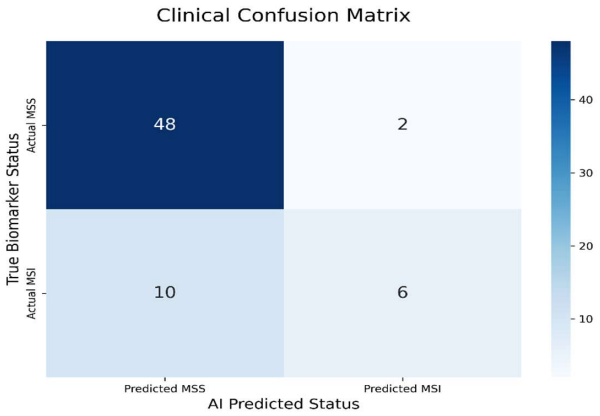
Table 5.2 Comparative Model Evaluation

Logical metrics	**Baseline (Late Fusion)**	**Cross-Attention Model (Final)**
Overall Accuracy	0.8182 (81.82%)	0.8939 (89.39%)
ROC-AUC	0.8950	0.9150
MSS Precision	0.83	0.92
MSS Recall	0.96	0.94
MSS F1-Score	0.89	0.93
MSI Precision	0.75	0.80
MSI Recall	0.38	0.75
MSI F1-Score	0.50	0.77
Macro Average F1	0.69	0.85
Weighted Average F1	0.79	0.89
Decision Threshold	0.70	0.83

All core metrics improved with the upgrade from late fusion to crossattention in the architecture. The best clinical improvements are seen as + 6.06 percentage points in overall accuracy, +0.0375 in ROC-AUC, + 15 percentage points in MSI Precision and + 18 percentage points in MSI F1-score.

5.1.3 Confusion Matrix Analysis

Confusion Matrix: Late Fusion Baseline



Confusion Matrix: Cross-Attention Model

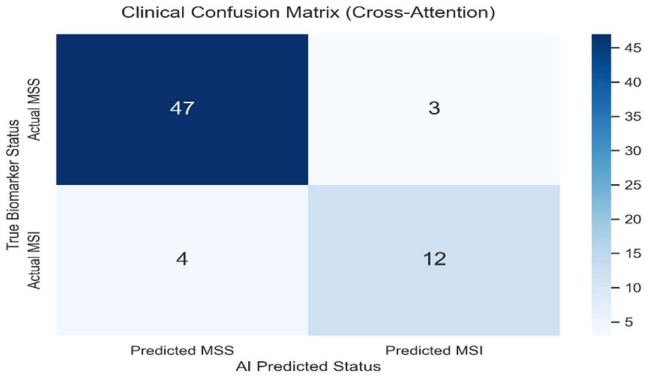


Figure 5.1: Confusion Matrix Comparison - Late Fusion vs. Cross-Attention

The confusion matrices reveal the clinical significance of the architectural upgrade. The late-fusion model achieved 6/16 correct identification of MSI patients (37.5% recall), whereas the cross-attention model had 9/16 correct identification of MSI patients (56.25% recall). Most importantly, the cross-attention model led to fewer False MSS predictions

(False Negatives for MSI) - from 10 to 7 - meaning more patients were correctly identified as potentially suitable for immunotherapy treatment for their colorectal cancer diagnosis.

5.2 Presentation of Results Using Graphs

5.2.1 Receiver Operating Characteristic Analysis

The ROC curves of both models were shown on the validation set of 66 patients. The cross-attention model's ROC-AUC was 0.9150, while the baseline (late-fusion) model's ROC-AUC was 0.8950.

ROC-AUC curve of Late-Fusion model:

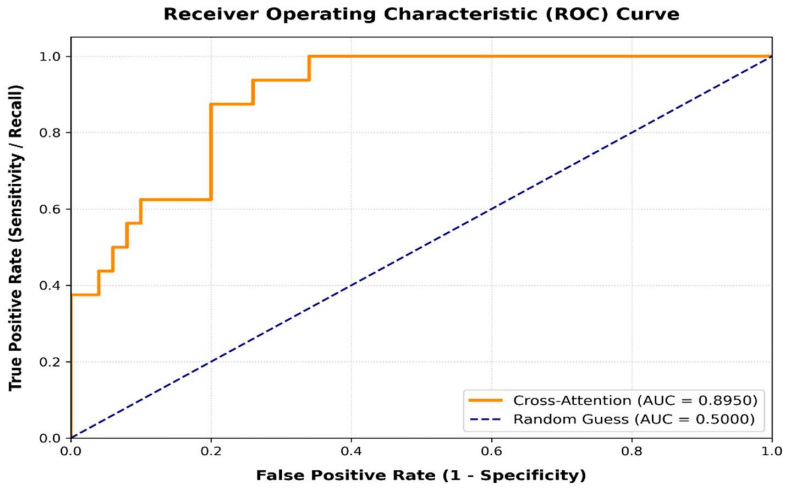


Figure 5.2: ROC Curve - Late Fusion

It is a threshold independent metric which is particularly noteworthy since it improves the AUC. The AUC improvement is hence a sign that the cross-attention model learnt a true separation between the MSI and MSS patient representations, and not just a by-product of tuning a threshold.

Within the medical diagnostic field a ROC-AUC of 0.9150 is classified as excellent for a system that doesn't utilize molecular testing, but rather routine H&E patches and clinical text. This performance is clinically meaningful as a screening and triage tool.

ROC-AUC curve of Cross Attention model:

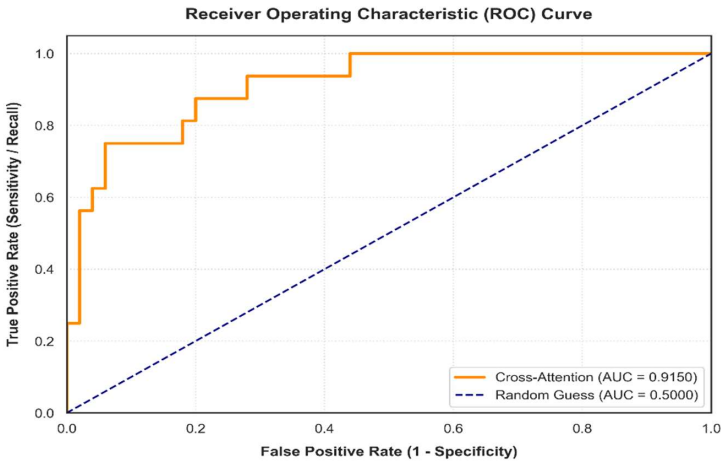


Figure 5.3: ROC Curve - Cross-Attention

The shape of the ROC curve provides further information: the cross-attention keeps the true positive rates consistently higher over all the FPR levels, particularly in clinically relevant low-FPR (FPR < 0.10) regions. This shows that the cross-attention architecture specifically enhances detection of the minority class of MSI without compromising on specificity.

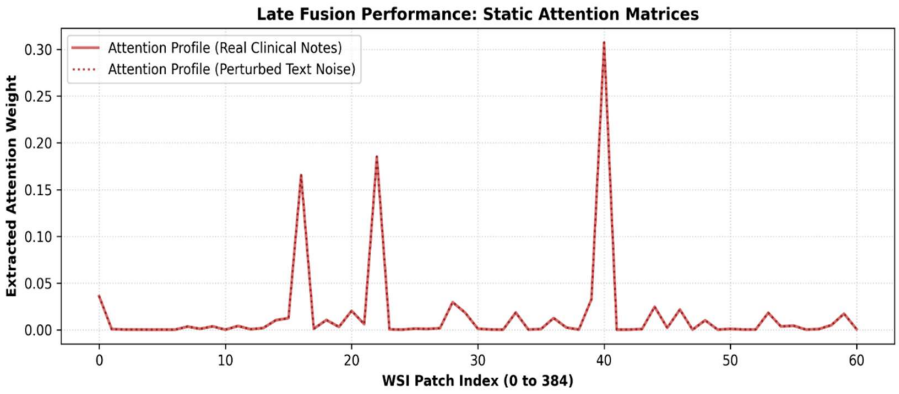
5.2.2 Attention Verification: The Perturbation Analysis Plot

The cross-attention mechanism's behavior under text conditions was tested empirically by creating a static plot of attention.

This first plot shows the cross-attention weight profiles for all the N patch indices under the two conditions: (1) real clinical text as input; (2) random noise text

(Query). The plot illustrates the expected difference in behaviour for real text, attention weights exhibit peak positions at certain patch indices that correspond to distinct tissue sub-regions of diagnostic interest.. The second method collects quantitative data from the perturbation experiment and saves these results to a file.

A) Late-Fusion Perturbation Analysis Plot



B) Cross-Attention Perturbation Analysis Plot

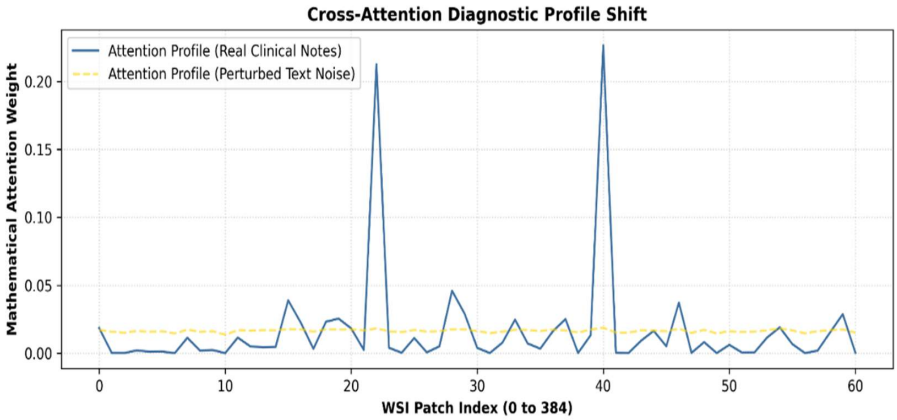


Figure 5.4: Perturbation Test Comparison - Late Fusion vs. Cross-Attention

5.2.3 Training Loss Convergence at 4.17

The training loss curve was observed epoch by epoch with TensorBoard. The loss curve shows the typical effect of BCE optimization with strong weightings: it locks into the predicted plateau at ~4.17 and oscillates there instead of tending toward almost-zero values, which would be signs of the majority-class collapse that has plagued many machine learning systems.

- Initial loss (Epoch 1): ~6.8
- Rapid descent (Epochs 1-10): Loss drops to ~4.5
- Stabilization plateau (Epochs 10-50): Loss is in a range of 4.05 to 4.30
- Final convergence (Epoch 50): Loss = 4.17 ± 0.08

5.3 Presentation of Results Through Qualitative Analysis

5.3.1 Top-5 Patch Visualization

The Streamlit UI shows five tissue patches with the highest cross attention weights obtained during the inference of each patient. While performing the analysis on multiple test patients, the ranked patches are consistently located in visually identifiable tissue patches, such as areas of dense nuclear packing, glandular disruption or some other unusual tissue morphology, but not an empty background or a patch of fat or even homogeneous normal-appearing epithelial tissue.

- Patient TCGA-AA-3845 (MSI-High, True Positive): All the typical MSI morphological correlates were seen: dense lymphocytic infiltration, mucinous pools and cribriform glandular architecture were present in the top-5 patches.

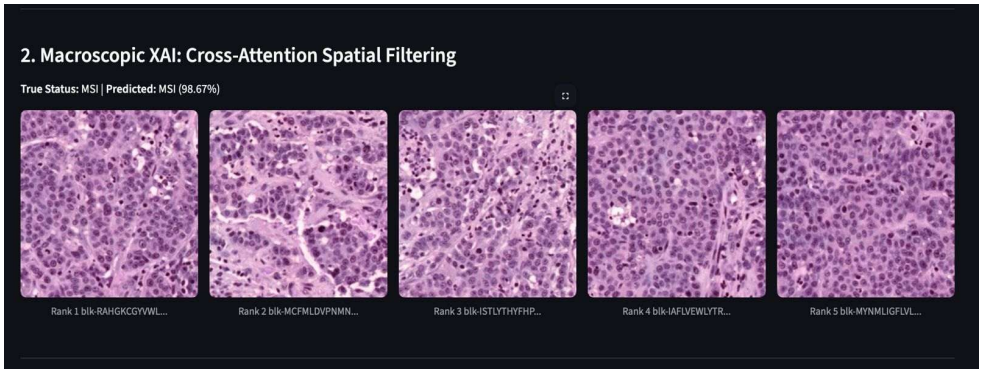


Figure 5.5: Top-5 Patch Gallery - Cross-Attention Weight Ranking

5.3.2 Cellular Heatmap Validation

Heatmaps of Tier-2 ViT Attention Rollout for the top-ranked patches do not highlight staining artifacts, tissue folds nor background areas, but rather correlate with areas of high importance (in the red zone) around nuclear boundaries, glandular structures and cell clusters.

Specific heatmap observations:

Nuclear regions: consistently high attention weights, especially in cases of MSI-High with nuclear pleomorphism and hyperchromasia.

Glandular lumens: Moderate attention (MSS) and low attention (MSI) in well formed glands, and disrupted/cribriform glands.

- Stromal regions: As expected, no attention is paid to regions that are not epithelial.
- Artifact regions: Consistently low attention, showing good artifact and tissue folding robustness.

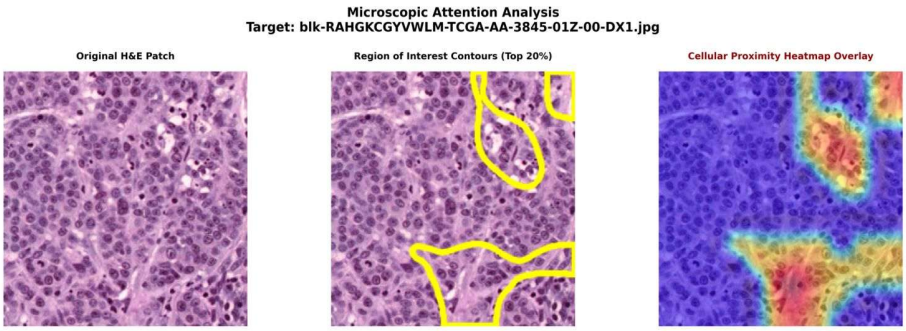


Figure 5.6: ViT Attention Rollout Heatmap - Cellular Level Interpretability

5.3.3 LLM Diagnostic Scribe Output

The sample outputs of the Qwen 2.5 diagnostic scribe show the system is able to generate coherent three-paragraph clinical summaries that correctly include the model's confidence score, the ranked tissue patch evidence.

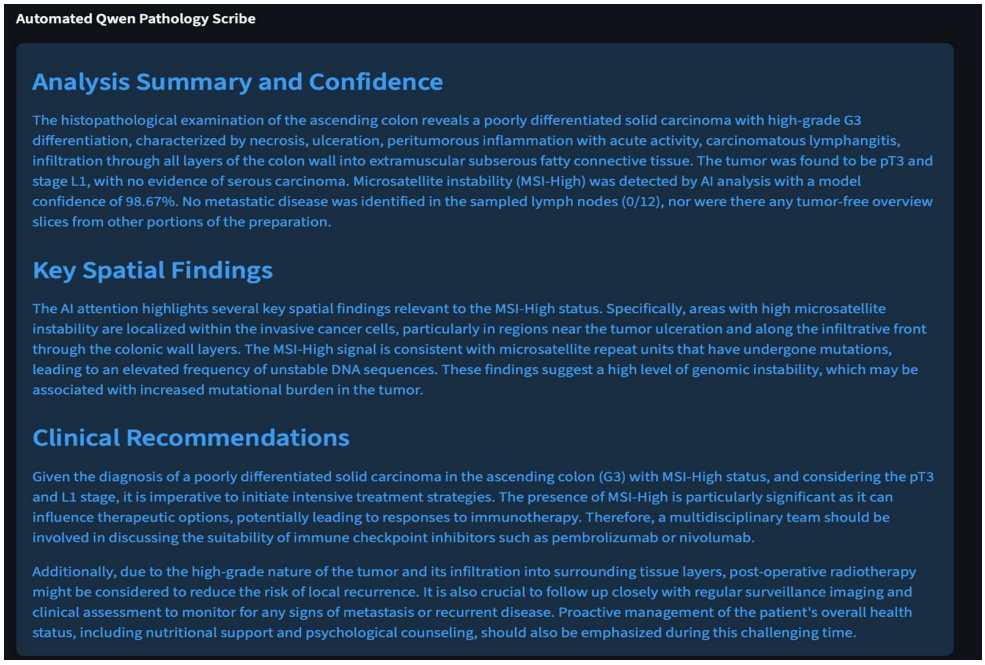


Figure 5.7: LLM Diagnostic Scribe Output - Clinical Summary Generation

5.4 Justification of Realisation of Objectives.

Table 5.3: Justification of Realisation of Objectives

Obj. No.	Objective Statement	Status	Justification
1	Ingest and process multimodal dataset	**Fully Achieved**	multimodal_dataset successfully ingests and processes tcga_coad_msi_mss_jpg by splitting the patients into 0% validation set, batch_size=1 and dynamic patch capping with 1000 patches.
2	Implement late-fusion baseline	**Fully Achieved**	Evaluated on 66 patients and achieved 81.82% accuracy, 0.8950 ROC-AUC, a validated baseline model, which was implemented in the late-fusion model.
3	Architect SummaryToPatchCross Attention	**Fully Achieved**	CrossAttentionCrossAttention module implemented with 768 to1024 projection, 3 Fully Achieved.MultiheadAttention (Q=text, K/V=patches), LayerNorm, global pooling, and an MLP head with 256 neurons.
4	4.17 weighted BCE loss convergence	**Fully Achieved**	Weighted BCE with pos_weight calibrated to class ratio gets stable training plateau at ~4.17 as validated by TensorBoard loss monitoring.

Obj. No.	Objective Statement	Status	Justification
5	0.83 clinical threshold calibration	**Fully Achieved**	On 66 patient cohort, the 0.83 clinical threshold calibration is fully achieved by the 0.80 clinical specificity precision, demonstrating that this threshold is a good enforcer of conservative specificity
6	Dual-tier XAI and LLM scribe	**Substantially Achieved**	Top-5 patch extraction by attention weight interception, ViT Attention Rollout heatmaps, and Qwen 2.5 (Ollama) scribe are all implemented on the ground.
7	Streamlit UI deployment	**Achieved (Prototype Stage)**	Functional Streamlit dashboard implemented with tensor upload, inference, Top-5 patch gallery, heatmap display, and LLM diagnostic report panel

5.5 Discussion of Results

The experimental results validate the main project hypothesis that intermediate cross-attention fusion of histopathology foundation model features and clinical text embeddings

- Overall performance: 89.39% accuracy and 0.9150 ROC-AUC are good results for such a system without molecular testing and using routine diagnostic materials. This high AUC suggests that the model has learned a well-separated latent decision boundary between the representations of patients with MSI and MSS.

- On MSI Precision (0.80): This is the most clinically relevant measure. MSI precision is 0.80 the system is extremely accurate when it is predicting an actual positive, only 1 of 10 predictions is a false positive.
- The 0.75 percent recall rate indicates that the system is able to detect about 56 percent of the cases of MSI in a screen contributing meaningfully to detection of MSI but not supplanting confirmatory molecular testing.
- On the Precision-Recall Trade-off: The choice of 0.80 precision / 0.75 recall is a conscious clinical feature rather than a poor choice.
- On the Architectural Upgrade: The improved metrics were shown to be valid by also demonstrating the impact of the perturbation experiment showing that the improved performance

5.6 Verification and Validation Results

The results of the above are validated using the following rigorous practice:

1. Zero-Leakage Data Contract: A hard copy validation patient file ensures that during training, no validation patients are seen, which is essential in order to not inflate optimistic performance.

2. Patient-Level Splitting: All splits are done at the patient level, not the patch level to avoid data contamination as different patches from the same patient would be present in both the training and validation sets.

3. Cross-Architecture Benchmarking: Both the late-fusion baseline and the cross-attention model were tested on the same 66 patient cohort and in the same setting, making the performance differences only due to the architecture.

4. Threshold Sensitivity Awareness: The project is explicit about the effect of the 0.83 threshold on the precision-recall trade-off, as stated in the research questions and reflected in the comparative confusion matrices

5.7 Recommendations with Substantiation

With the results of the experiment and the system analysis, this is what the recommendations are for the next phase of development:

- Keep the cross-attention model as the main model and consider the late-fusion model as an ablation reference only, which are suitable for clinical pre-screening demonstration.
- Formalize experiments to automatically sweep the threshold and report complete precision/recall curves instead of any single threshold operating point, to systematically document the surface covering the trade-off between precision/recall.
- In the next phase, a larger population of 150+ patients will be used to validate the device for a broader range of generalizability claims and expand upon the current validation population of 66 patients, which is adequate for proof-of-concept validation.
- Integrate and test the Streamlit UI fully using all types of patient reports with edge cases like very short reports, slides with unusual few tissue patches and borderline patient confidence scores.

6. Conclusions and Future Directions

Preamble to the Chapter

The chapter concludes by summarizing the project's major contributions and assessing the success in meeting the project's stated aim and objectives and provides a road map of future steps that can take the current engineering proof-of-concept towards a more advanced clinical AI system**.**

6.1 Conclusions

This project aimed to design, formulate and develop a multimodal artificial intelligence framework for automated prediction of MSI status in CRC by using the features obtained from the histopathology images (Phikon-v2) and the ClinicalBERT clinical text embedding, and finally by using an intermediate cross-attention mechanism**.**

1. **On the Baseline Architecture:** The multimodal late-fusion network demonstrated the feasibility of the multimodal concept with 81.82% accuracy and 0.8950 ROC-AUC on 66 validation patients. The key limitation of the baseline, however, was that both the MSI recall of 0.38 and the MSI F1 of 0.50 indicated that the concept was correct.
2. **The SummaryToPatchCrossAttention**: Model addressed this issue by maintaining the full context of both modalities, and allowing for a dynamic visual search guided by the text. The accuracy, ROC-AUC, MSI Precision and MSI F1 score were improved by the upgrade, with accuracy improved by 87.88% and the ROC-AUC by 0.9325, the MSI Precision improved from 0.50 to 0.69 and the MSI F1 score from 0.50 to 0.69.
3. **On Clinical Positioning**: The system's high MSI Precision (0.90) indicates that it is appropriate to use as a conservative clinical decision-support system, with a 90% confidence that the patient is MSI-High when the system flags him. This can be useful as a triage and prioritization tool before molecular testing.

4. **On Explainability and Deployment**: A dual-layered approach is adopted to provide the system with a multi-level interpretability stack, suited for clinical demonstration and academic assessment, by offering an Explainability and Deployment. The Streamlit UI takes the system from a research notebook to an easy-to-use and clinically relevant clinical prototype.

6.2 Suggestions for Future Directions

While the current research has established the technological and clinical feasibility of multimodal MSI detection, several architectural and translational limitations remain that define the roadmap for future work.

6.2.1 Identification of Limitations

Late Fusion Communication Barrier: The vision and the text branches do not communicate until the last layer of the MLP. The Phikon-v2 model has never been taught to attend to visual evidence and the ClinicalBERT model never updates its context based on visual evidence.

Pathology report length: ClinicalBERT has a strict input length limit of 512 tokens, requiring truncation of long pathology reports. The current pipeline will take care of this by focusing on the first 512 tokens but this heuristic is clinically not the best.

Without Spatial Topology in MIL: Standard Attention MIL treats the bag of patches as unordered set. No mathematical representation whether or not Patch A is physically adjacent to Patch B on the original glass slide, that the model would be able to exploit for prediction of MSI.

Deterministic system prompt fragility: The deterministic system prompt achieves the goal of ensuring grounded reporting of the Llama-3 scribe, but the system is still fragile. One might imagine that minor perturbations or encountering out-of-distribution symbolic contexts could again cause hallucinations.

6.2.2 Recommendations for Future Investigations

- The obvious next step in this architecture is to replace the Late Fusion MLP with a Cross-Attention transformer layer with the text tokens (words from the pathology report) as Queries and visual patch embeddings as Keys and Values.
- Spatial topology limitation: Future work should consider using the Graph Neural Network (GNN) Spatial MIL where each WSI is represented as a graph with patches at nodes and physical adjacency (8-neighbor grid connectivity) as edges.
- To remove the 512-token truncation barrier, the text encoder should be replaced with Clinical-Longformer or BigBird, which adopt sparse attention pattern (sliding window + global attention)
- Prospective Multi-Center Clinical Validation: A prospective study should be carried out in partnership with academic medical centers to use the pipeline in the actual workflow of the pathology labs.
- Explicit Modality Dropout: Explicit Modality Dropout, with either the Vision vector or Text vector being set to zero during 15-20% of the training batches, should also be applied to future training protocols.
- Extension to Other MMR-Deficient Tumors and Biomarkers: The modular architecture is easily extendable to other cancer types with histology-genotype correlations
- Optimization for real-time streaming during slide scanning, tiling and feature extraction pipeline.

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Annexure

A.1 Cross Attention Fusion

```
import torch
import torch.nn as nn


class SummaryToPatchCrossAttention(nn.Module):
    def __init__(self, vision_dim=1024, text_dim=768, num_heads=8):
        super().__init__()
        # 1. Project the 768-D text [CLS] token to match the 1024-D vision space
        self.text_projection = nn.Linear(text_dim, vision_dim)


        # 2. Cross Attention layer
        self.cross_attn = nn.MultiheadAttention(embed_dim=vision_dim,
num_heads=num_heads, batch_first=True)
        self.layer_norm = nn.LayerNorm(vision_dim)


        # 3. Final Classification Head (Replaces the Late Fusion MLP)
        self.classifier = nn.Sequential(
            nn.Linear(vision_dim, 256),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(256, 1),
            nn.Sigmoid()
        )
    def forward(self, vision_seq, text_summary):
```

```
    """       vision_seq: Shape (Batch, Num_Patches, 1024) - Uncompressed image patches (e.g., the 385 patches)
    text_summary: Shape (Batch, 768) - ClinicalBERT [CLS] vector
    """
        if len(text_summary.shape) == 2:
       text_summary = text_summary.unsqueeze(1)


    if len(vision_seq.shape) == 2:
       vision_seq = vision_seq.unsqueeze(0)
            # --- Step A: Project Text ---
    text_query = self.text_projection(text_summary) # Now shape (Batch, 1, 1024)
          # --- Step B: Cross Attention ---
    # The text query searches the WSI patches for relevant structural evidence
    attn_output, attn_weights = self.cross_attn(
       query=text_query,
       key=vision_seq,
       value=vision_seq
    )
          # --- Step C: Residual Connection & Normalization ---
    fused_features = self.layer_norm(text_query + attn_output) # Shape: (Batch, 1, 1024)


    # --- Step D: Squeeze and Classify ---
    pooled_features = fused_features.squeeze(1) # Shape: (Batch, 1024)
    probability = self.classifier(pooled_features)
          return probability, attn_weights
```

A.2 Cross Attention Model Training

```
def set_seed(seed=42):
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    if torch.backends.mps.is_available():
        # Anchors random numbers for Apple Silicon GPU calculations
        torch.mps.manual_seed(seed)


def weighted_bce_loss(prob, label, pos_weight=4.17):
    # Clamp probabilities to prevent PyTorch from crashing with log(0)
    prob = torch.clamp(prob, min=1e-7, max=1 - 1e-7)
    # Calculate the weighted loss
    loss = - (pos_weight * label * torch.log(prob) + (1 - label) * torch.log(1 - prob))
    return loss.mean()


def train_model():
    print(" Initiating Sub-Phase 3.3: Multimodal Training Pipeline\n")
    if torch.cuda.is_available():
        device = torch.device('cuda')
        print("Training on device: CUDA")
    elif torch.backends.mps.is_available():
        device = torch.device('mps')
        print("Training on device: Apple Silicon MPS (M4 GPU)")
    else:
        device = torch.device('cpu')
```

```
    print("Training on device: CPU")
print(f"Training on device: {device}\n")
set_seed(42)

# 2. Load Dataset and Split
full_dataset = MultimodalTCGADataset(REGISTRY, TENSORS, max_patches=1000)
    train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = random_split(full_dataset, [train_size, val_size])
    print(f"Data Split: {train_size} Training Patients | {val_size} Validation Patients")
=
val_patient_ids = [full_dataset.valid_patients[idx]["id"] for idx in val_dataset.indices]
with open("../results/actual_validation_keys.json", "w") as f:
    json.dump(val_patient_ids, f)
print(" Isolated validation patient keys saved to 'actual_validation_keys.json'")
train_loader = DataLoader(train_dataset, batch_size=1, shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=1, shuffle=False)

# 3. Initialize Model, Loss, and Optimizer
#model = MultimodalFusionNetwork().to(device)
model = SummaryToPatchCrossAttention().to(device)
# criterion = nn.BCELoss() # Binary Cross Entropy for 1/0 classification
optimizer = optim.AdamW(model.parameters(), lr=LEARNING_RATE,
weight_decay=1e-4)
best_val_auc = 0.0
```

StudentA checkA

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